

Endogenous Supplier Sourcing under Delivery Risk and its Macro Implications

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Abstract

This paper studies how delivery risk - fluctuations in waiting time due to port congestion - shapes firms supplier sourcing decisions and the resulting aggregate outcomes. Using granular transaction-level trade data for U.S. firms, I show that higher delivery risk induces firms to adopt precautionary sourcing strategies, diversifying their supplier base and shifting toward safer suppliers to stabilize production. While such reorganization reduces output volatility, it raises production costs, generating a fundamental trade-off between stability and efficiency.

To rationalize these findings and quantify their aggregate implications, I develop a general equilibrium model with endogenous supplier sourcing and delivery risk. In the model, the mere possibility of delivery delays - independent of realized disruptions - induces firms to engage in costly precautionary behavior. As a result, equilibrium prices are higher and aggregate output is lower, even when no actual disruptions occur. The framework provides a mechanism consistent with the persistence of elevated prices and subdued output following the acute phase of the COVID-19 pandemic. More broadly, the results highlight how supply-chain uncertainty can generate allocative inefficiencies through defensive firm behavior, suggesting that policies aimed at reducing uncertainty may yield larger welfare gains than policies that focus solely on mitigating realized disruptions.

Keywords: supplier sourcing, delivery risk, supply chains

JEL Classification: D22, F14

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1 Introduction

Over the past two decades, global supply chains have been repeatedly exposed to large disruptions that not only lengthen delivery times but also increase uncertainty in delivery performance. Figure 1 plots the ISM Manufacturing Suppliers Delivery Time Index alongside realized delivery risk, defined as the rolling standard deviation of the index over the past 12 months. The figure highlights pronounced episodes of delivery delays associated with major geopolitical, natural, and economic events, including the Iraq War in 2003, the 2011 earthquake and tsunami in Japan, the U.S.China trade tensions in 2018, and the COVID-19 pandemic in 2020. Importantly, these episodes are characterized not only by elevated delivery times but also by heightened volatility and frequent fluctuations in delivery conditions, as reflected in the close comovement between the level index and the delivery risk measure, indicating increased unpredictability in input arrivals.* For firms that rely on timely intermediate inputs, such variability in delivery performance poses a large source of risk. Even when average delivery times decline, greater uncertainty about whether and when inputs will arrive can disrupt production planning and amplify output volatility. As climate-related shocks, geopolitical conflicts, and trade policy uncertainty become more frequent, delivery risk is likely to persist as a central feature of the production environment.

Such delivery risk is costly. In a recent paper Alessandria et al. 2023 , they estimate that the delay in importing goods lead to a substantial contraction in economic activity, with production, consumption and employment falling sharply on impact and at the same time increase price.

Though delivery risk impact is large, firms may mitigate this risk by reducing their reliance on risky suppliers and opt for new suppliers who are safer to diversify the risk. In a recent survey[†], it documents a potentially drastic shift in the way manufacturing company executives generally think about their supply chains: as a result of covid, 70% company agreed that they will focus less on sourcing from the lowest-cost suppliers and instead prioritizes stability and resilience in the face of disruption, 40% of respondents are planning on or already doing multi-sourcing instead of reply on a single source for the supply of various. Those response reflect firms concerning that the loss of sales during disruptions may

*The ISM Manufacturing Supplier Deliveries Index is a key macroeconomic indicator that measures the pace of delivery of materials from suppliers to manufacturers within the United States. The index is compiled and reported by the Institute for Supply Management (ISM) on a monthly basis, based on surveys of purchasing and supply executives nationwide. High Supplier Deliveries Index suggests that suppliers are facing delays in delivering materials to manufacturers, potentially signaling strong demand and constrained production levels in the manufacturing sector. Conversely, a low index value indicates quick deliveries, which may reflect weaker demand or efficiency improvements in the supply chain.

[†]Survey by Foley & Lardner LLP, available online at <https://www.foley.com/wp-content/uploads/imported/9e39e02aa5fa4110b4354cbe3b13cfee.pdf#page=6.13>

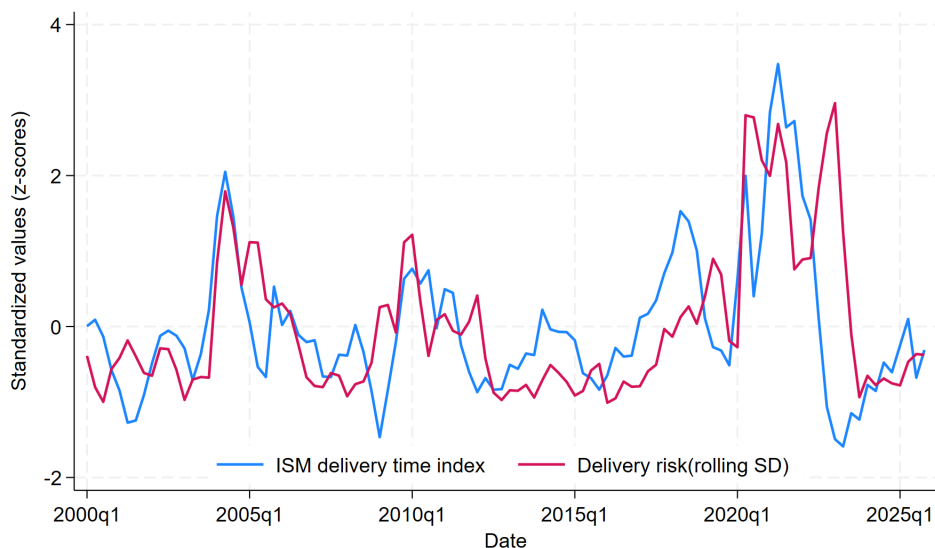


Figure 1: Level and Rolling Standard Deviation of ISM Delivery Time Index

Notes: The figure plots the ISM manufacturing delivery time index and its realized delivery risk, measured as the standard deviation computed over a rolling 12-month window. The series are shown at quarterly frequency from 2000 to 2025, and both the level index and the rolling standard deviation are standardized over the 2000-2025 sample period. A value of 2 for the delivery risk measure indicates that realized delivery risk is two standard deviations above its historical average.

outweighs keeping the cost of manufacturing down.

This paper analyzes how firms adjust their supplier sourcing in response to delivery risk and the resulting aggregate and macroeconomic implications. Using granular transaction-level trade data on U.S. importers, I document several key empirical patterns that characterize the underlying mechanism. First, firms respond to elevated delivery risk by diversifying their supplier base. Second, beyond diversification among similar suppliers, firms exhibit risk-shifting behavior, reallocating sourcing away from suppliers located in higher-risk countries toward those in lower-risk locations. Finally, I show that delivery-risk-induced diversification generates a fundamental trade-off between efficiency and stability: while it reduces volatility, it increases production costs and lowers firms' profit margins and markups. These firm-level sourcing responses and outcomes then transmit to aggregate economic outcomes.

Motivated by empirical evidence, I develop a general equilibrium model with endogenous supplier sourcing to study the role of delivery risk in firms' sourcing strategies and its aggregate consequences. In the model, firms form beliefs about their suppliers' delivery reliability and choose which suppliers to engage with, how much to source from each, and the scale of production to maximize expected utility. The model shows that, in the presence of delivery risk, firms prioritize stability by diversifying their supplier base, often shifting

toward safer but less productive partners. This risk diversification strategy stabilizes output but raises production costs, leading to higher aggregate prices and lower aggregate income. Importantly, these effects arise not only when delivery disruptions are realized but also in their absence, as firms adopt precautionary sourcing strategies to safeguard supply chains and output. This mechanism helps explain why elevated prices can persist even after the resolution of large disruptions such as the COVID-19 pandemic, as continued uncertainty sustains precautionary behavior. From a policy perspective, improving delivery reliability can yield aggregate welfare gains not only by mitigating the impact of actual disruptions but also by reducing inefficiencies driven by persistent uncertainty.

2 Contribution to Literature

Our study contributes to the rapidly growing micro-oriented literature that examines the effects of supply disruptions on firms with input-output linkages and considers their aggregate implications. For instance, Carvalho et al. 2021 analyze the firm-level impact of the Thoku earthquake and quantify its aggregate effects in a closed economy general equilibrium model with production linkages. Barrot and Sauvagnat 2016 demonstrate that natural disasters affecting upstream suppliers can have large and persistent effects on the value and production of downstream firms.

However, most existing papers do not address the supply constraints related to delivery risk, which we aim to emphasize. Our paper seeks to incorporate this increasingly important type of shock heightened delivery times and risks into macroeconomic models for business cycle analysis.

In this context, our work aligns closely with Alessandria et al. 2023, which quantifies the aggregate effects of supply chain delays and depleted inventories experienced in the United States and globally in the post-COVID period. Building on their paper, our paper distinguishes itself by introducing an additional margin of analysis that focuses on the endogenous selection of suppliers in response to delivery risks. Our empirical analysis provides strong evidence that the number of suppliers is positively correlated with delivery risk, even when using data from before COVID-19. This underscores that the extensive margin adjustment in supplier sourcing due to delivery risks is not confined to the pandemic; rather, it has broader implications for supply chain management across various contexts.

Another closely related paper is Castro-Vincenzi et al. 2024 which develops a model of firm input sourcing under climate risk in India, demonstrating how firms diversify their inputs from suppliers across regions, balancing the probability of climate shocks against higher input costs and their implications for local wages. While *ibid.* focuses on climate-

related production risk, this paper instead studies delivery risk arising from fluctuations in waiting times due to port congestion. Moreover, whereas Castro (2024) emphasizes regional labor market effects, this paper examines how precautionary supplier adjustments transmit to aggregate economy, generating an efficiency-stability trade-off at the macroeconomic level.

Our work also contributes to the literature on the endogenous formation of production networks in the context of macroeconomic uncertainty, as well as to the trade literature on firms' input choice behavior.

Kopytov et al. 2024 constructs a model of endogenous network formation to investigate how these decisions influence the structure of production networks and the levels and volatility of macroeconomic aggregates. In their work, firms choose production techniques in the presence of uncertainty and explore how links between sectors are created or destroyed. This is examining how firms substitute inputs across industries with varying uncertainties. However, substitution across industries often takes time, and in many cases, firms' multi-sourcing strategies focus on finding additional suppliers for even single identical input. As a result, our work examines how firms adapt their sourcing strategies across suppliers, whether within a single industry or across different industries and input types. Additionally, while *ibid.* work focuses on uncertainty shocks in terms of the variance of productivity, we investigate delivery risk, which encompasses the probability of delivery failure or the fluctuation of delivery time factors that are becoming increasingly significant in today's context.

There is also a strand of trade literature that addresses firms' endogenous input choices. For example, Handley et al. 2024 examines the role of trade policy uncertainty in shaping firms' import decisions. Dhyne et al. 2023 develops a model that includes the endogenous formation of links between firms in response to trade shocks. However, their focus is primarily on trade-related shocks, such as China's accession to the WTO, trade policy uncertainty, and rising import prices, and their implications for long-term trends. In contrast, our research centers on delivery risk, which is becoming increasingly relevant in today's world marked with more frequent climate shocks and geopolitical conflicts, and aims to investigate its implications even for the business cycle in short run.

The remainder of the paper is organized as follows. Section 3 presents empirical evidence along several dimensions, documenting how U.S. firms adjust their supply-chain sourcing behavior in response to delivery risk and how changes in supplier relationships affect production costs and volatility. Section 4 develops a general equilibrium model with endogenous supplier sourcing to study firms' diversification responses to delivery risk and their implications for aggregate output and prices.

3 Empirical Analysis

In this section, we describe the data used in the analysis and present empirical evidence on how firms adjust their supplier sourcing in response to delivery risk and the resulting effects on firm outcomes.

First, we document that higher delivery risk induces U.S. importers to diversify their supplier base. Second, beyond pure diversification, we find clear evidence of risk-shifting behavior: U.S. importers reallocate sourcing away from countries with higher delivery risk toward countries with lower delivery risk. Third, we show that delivery-risk-induced diversification reduces firm-level volatility but increases production costs and compresses profitability, a fundamental trade-off between efficiency and stability.

3.1 Data and Variable Construction

3.1.1 Data overview

Panjiva The primary dataset used in this paper is S&P Panjiva, which is a comprehensive bill of lading (BoL) database encompassing more than one billion shipment-level records of cross-border trade transactions. The data includes both import and export transactions for around 20 countries worldwide.[‡] This dataset has become increasingly popular in recent trade and supply-chain research due to its detailed coverage of firm-to-firm international transactions.

A bill of lading is a legal document that serves as evidence that a shipment has been transported from its origin to its final destination. While Panjiva provides shipment-level data for approximately 20 countries, this paper focuses on U.S. imports. For the United States, S&P Global Panjiva obtains bill-of-lading information from U.S. Customs and Border Protection (CBP) through the Freedom of Information Act (FOIA) of 1966. The U.S. BoL data include nearly 200 million records and span the period since 2007.

The bill-of-lading data describe the contractual relationship between the shipper and the consignee. Firms are required to complete a variety of fields in each bill of lading, including the names and addresses of the shipper (exporter) and consignee (importer), descriptions of the goods, vessel name, transport company name, ports of lading (loading) and unloading (unloading), as well as shipment weight and container information. Additionally, the dataset includes supplementary variables imputed by Panjiva itself, such as shipment volume measured in twenty-foot equivalent units (TEUs), Harmonized System (HS) product codes at the six-digit level for each shipment, and - in a limited subset of cases - an estimated

[‡]

transaction value in U.S. dollars. Crucially, the data identify both parties involved in each transaction, namely the consignee (U.S. importer) and the shipper (foreign seller), each of which is assigned a unique company identifier. Panjiva further provides a cross-reference file that allows trade transactions for some shippers and consignees to be linked to their corresponding firms in other datasets, such as S&P Capital IQ, Compustat, and Worldscope. In this paper, we merge the import transaction data with U.S. Compustat to obtain firm-level financial and operational information for importing firms. Table 1 reports the variable names and descriptions for selected key variables in the Panjiva BoL data, including both raw and imputed variables.

Table 1: US import data description for selected variables

Variable	Description
panjivarecordid	Unique shipment ID
arrivaldate	Day of arrival
vessel	Name of the vessel that transported the goods
portoflading	Port of lading
portofunlading	Port of unloading
weightkg	Shipment weight in kilograms
volumeteu	Volume of shipment in TEUs
ValueofgoodsUSD	Value of goods in USD
conpanjivaid	Consignee ID
shppanjivaid	Shipper ID
hscode	6-digit HS code
companyid	Capital IQ company ID
shpmtorigin	Location from which shipment left for the US

Congestion Time Index In addition to shipment-level trade data from S&P Panjiva, we use a monthly port-level congestion time index constructed by Bai et al. 2024 to measure delivery delays and supply-chain disruptions. The congestion time index is built using satellite-based data from the Automatic Identification System (AIS) - a vessel tracking system mandated by the International Maritime Organization (IMO) - which records the position, speed, and heading of container ships. This enables measurement of congestion at major ports around the world. The congestion time index measures the average number of hours that a container ship waits in a ports anchorage area before docking at a berth to load or unload cargo. Port congestion time can arise from upstream and downstream frictions along the supply chain, including reduced port-handling efficiency due to health and safety protocols, delays in inland transportation that prevent timely container pickup, and containers remaining unopened at inland facilities because of labor shortages. These frictions propagate through the logistics

network and accumulate at ports, ultimately manifesting as prolonged vessel waiting times.

Port congestion waiting time serves as a natural and direct measure of delivery delays for seaborne trade. First, congestion mechanically increases transit time and elevates uncertainty regarding the arrival of imported inputs. Second, containerized maritime shipping plays a central role in international trade - more than half of global trade flows are conducted via maritime shipment - and container ports serve as critical hubs in tightly scheduled logistics networks. When congestion emerges at these hubs, even modest delays can propagate through shipping routes and generate substantial disruptions to downstream production and inventory management. This measure is particularly well-suited to our empirical setting, as U.S. import transactions observed in the Panjiva data are predominantly transported by container ships and therefore necessarily transit through maritime ports.

A key advantage of port congestion time relative to commonly used survey-based delivery measures such as the Purchasing Managers' Index (PMI) or Institute for Supply Management (ISM) supplier delivery indices is that it is relatively more exogenous. Container shipping operates on largely fixed routes under binding service agreements that are predetermined and costly to adjust in the short run. Consequently, port congestion is only weakly influenced by the strategic decisions of individual shippers or by reallocations of capacity across routes in response to demand fluctuations. Bai et al. 2024 argues that, due to this institutional structure, port congestion times are relatively exogenous to demand-side forces that might otherwise confound measures of supply chain disruptions or reflect managers' subjective assessments of supply chain conditions.

Finally, port congestion increased sharply and remained elevated during the COVID-19 pandemic, closely aligning with widely documented episodes of global supply-chain stress. This sharp and persistent rise in waiting times provides additional validation that the congestion time index captures economically meaningful delivery delays associated with supply-chain disruptions. The index covers 47 major ports across 24 countries over the period from 2016 to 2025, with these 24 countries accounting for approximately 75 percent of trade transactions and shipments recorded in the Panjiva database.

We use the congestion time index as the primary input to construct our time-varying, firm-level delivery risk measures, which we describe in detail below.

3.1.2 Sample Construction

Our sample construction begins with the complete set of shipments received by U.S. consignees. Initially, we exclude observations lacking panjiva firm identifiers, particularly those with missing *conpanjivaid*. Subsequently, we employ a cross-reference file to connect each *conpanjivaid* with its corresponding *companyid*, which serves as the firm identifier in S&P

Capital IQ. Not every consignee can be successfully linked, as numerous firms are privately held and too small for Capital IQ coverage due to scarce publicly available information, even though they participate in international trade. We therefore exclude observations with missing *companyid* from our sample. In our analysis, we work at the level of the ultimate parent US company so we aggregate the data at the ultimate parent company level. We accomplish this by using the S&P Global cross-reference file to associate each *companyid* with its ultimate parent identifier (*ultimateparentcompanyid*). We remove observations lacking ultimate parent identifiers, though this criterion eliminates only a minimal number of shipments.

Finally, our analysis focuses on the subset of consignees that can be linked to the universe of Compustat firms. We use a crosswalk file obtained from WRDS to merge the Panjiva data with Compustat. This crosswalk provides the start and end dates for each ultimate parent company Capital IQ IDGVKEY linkage, and we merge with the link that is active in a given period. Our sample spans 2016:2025, corresponding to the availability of the port congestion time index constructed by Bai et al. 2024 and covering both the pre- and post-COVID periods.

Table 2 reports summary statistics for 2019 for two samples: the full sample of U.S. importers with identified ultimate parent Capital IQ firm IDs, and the subsample that can be merged with Compustat, which primarily consists of larger firms. The full sample includes approximately 130,000 ultimate parent firms, while the Compustat-linked sample contains about 7,000 large firms. Across both samples, firm-level import activity is highly right-skewed in terms of the number of shipments, import volume, and the number of shippers. For instance, in the Compustat sample, the average firm conducts roughly 500 import transactions per year, corresponding to about 1,300 containers measured in twenty-foot equivalent units (TEUs). The average firm sources from around 20 distinct shippers annually, whereas the median firm engages with only four. Firms in the top decile of the distribution import from more than 50 foreign suppliers. The bigger sample exhibits broadly the same patterns as the Compustat public firms sample.

3.1.3 Construction delivery risk measurement

The congestion time index constructed by *ibid.* provides a direct measure of delivery delays arising from port congestion and thus captures the first moment of shipping conditions. The central objective of this section, however, is to quantify delivery risk in addition to realized delivery delay, at the firm level. To this end, we construct a shiftshare measure of delivery risk that captures firms exposure to fluctuations in port congestion time. Specifically, we measure firms exposure to shipping time risk using the volatility of the port congestion time index faced by U.S. importers. This firm-level delivery risk measure allows us to examine

Table 2: Summary Statistics, 2019

Variable	Number of Firms	Mean	Std. Dev.	P10	P50	P90
Full Sample						
Shipments	134,508	56.487	1,470.631	1.000	4.000	78.000
Volume (1000s TEUs)	134,508	140.972	3,121.959	0.100	4.620	147.520
Number of Shippers	134,508	5.233	16.256	1.000	2.000	11.000
Value (million USD)	134,508	10.937	252.812	0.005	0.205	7.246
Compustat						
Shipments	7,014	487.665	6,417.297	1.000	18.000	669.000
Volume (1000s TEUs)	7,014	1,278.534	13,731.060	0.370	26.230	1,523.290
Number of Shippers	7,014	21.771	69.384	1.000	4.000	52.000
Value (million USD)	7,014	116.975	976.069	0.018	1.489	115.504

whether and how firms adopt precautionary sourcing strategies in response to heightened delivery risk. We begin by collapsing the monthly port-level congestion measures to a monthly country-level panel, aggregating across major ports within each country. This aggregation reflects the assumption that congestion at a country's major container ports is representative of overall port congestion conditions faced by shipments entering or leaving that country.

We then propose a three-step approach to construct the delivery risk.

The first step is to isolate unexpected fluctuations in port congestion by residualizing the raw congestion indices. Specifically, we estimate regressions of congestion index on country fixed effects, month-of-year fixed effects, and a linear time trend. Specifically,

$$\text{Congestion}_{ct} = \alpha_c + \lambda_{m(t)} + \beta \text{trend} + \varepsilon_{ct}, \quad (1)$$

where α_c denotes country fixed effects, $\lambda_{m(t)}$ are month-of-year fixed effects, and trend is a linear time trend. Country fixed effects α_c absorb time-invariant differences in port infrastructure, geography, distance from US and baseline congestion levels across countries; month-of-year fixed effects $\lambda_{m(t)}$ capture systematic seasonal patterns in shipping activity; and the time trend removes common global trends in congestion over the sample period. The residual from this regression therefore represents deviations in port congestion that are unexpected given a country's typical congestion level, seasonal shipping patterns, and long-run global trends. Economically, the residualized congestion index captures high-frequency, within-country innovations to port congestion that are plausibly driven by unanticipated disruptions rather than predictable seasonality or persistent structural differences across ports. This residualization relies on the assumption that the included fixed effects and trend adequately capture the predictable component of congestion, so that the remaining variation reflects unexpected delivery frictions. By focusing on this residual variation, we abstract

from slow-moving or systematic factors and instead isolate the component of port congestion most relevant for measuring time-varying delivery risk faced by importing firms.

In the second step, we quantify delivery risk by measuring the volatility of the residualized port congestion index. For each country and month, we compute rolling standard deviations of the residualized congestion measures over six-month and twelve-month windows. These rolling volatilities capture time-varying uncertainty in delivery conditions, abstracting from predictable components of congestion and focusing instead on fluctuations that are unexpected and persistent in maritime logistics.

In the third step, we aggregate country-level congestion volatility to the firm level using firms import sourcing patterns. Specifically, for each firm f and period t , we construct a weighted average of congestion volatility across source countries, where the weights reflect firm f 's import shares from each country. In the baseline specification, we use fixed import shares computed prior to 2016, before the start of the congestion index. For each firm, these shares are calculated based on containerized import volumes during the period 2013 - 2015 and are held constant over time. ie.

$$\widehat{\text{DeliveryRisk}}_{f,t}^{(h)} = \sum_c w_{f,c}^{\text{pre}} \sigma_{c,t}^{(h)}, \quad (2)$$

where the weights $w_{f,c}^{\text{pre}}$ are firm f 's pre-2016 import shares from source country c , and the country-level standard deviations $\sigma_{c,t}^{(h)}$ capture time-varying volatility in port congestion. Weighting country-level congestion volatility by these pre-period shares yields a firm-level delivery risk measure that takes the form of a shiftshare variable: firms that source more heavily from countries experiencing greater volatility in port congestion are more exposed to delivery risk, while firms with more diversified or less exposed sourcing patterns face lower risk. By construction, this approach ensures that firms exposure weights are predetermined with respect to subsequent congestion shocks, isolating variation driven by changes in delivery risk rather than endogenous adjustments in sourcing. Finally, we aggregate the resulting monthly firm-level delivery risk measures to the quarterly frequency by taking within-quarter averages.

3.2 Case Example of Ford

Our study is motivated by direct observations of firms extensive- and intensive-margin adjustments in the data, prior to conducting formal empirical specifications and structural mod-

eling. An illustrative example is provided by Ford. Following the onset of COVID-19 when delivery risk became particularly salient Ford exhibits pronounced diversification at the extensive margin. Figure 2 shows a sharp and persistent increase in the number of suppliers and supplier ultimate parent companies, indicating a broadening of its sourcing network. In contrast, we do not observe a comparable structural shift at the intensive margin for Ford: measures of shipment volume, including the number of containers, TEUs, shipment weight, and shipment value, display no sustained upward trend over the same period.

Beyond this structural shift at the extensive margin, the cyclical component of the number of suppliers -obtained using an HP filter - exhibits pronounced comovement with delivery risk: Figure 4 shows that periods of elevated delivery risk are associated with larger short-run increases in the number of suppliers, a pattern that becomes markedly stronger after 2020. Importantly, increases in supplier diversification for Ford tend to precede peaks in the realized delivery-risk measure, consistent with a precautionary, forward-looking adjustment in supplier sourcing in response to delivery risk.

This comovement is not unique to Ford. Figure 5 plots the average number of suppliers against the average delivery risk across all firms in the sample over time and reveals a similar pattern: periods of elevated delivery risk are associated with higher average supplier counts, indicating systematic comovement between delivery risk and supplier diversification at the aggregate level.

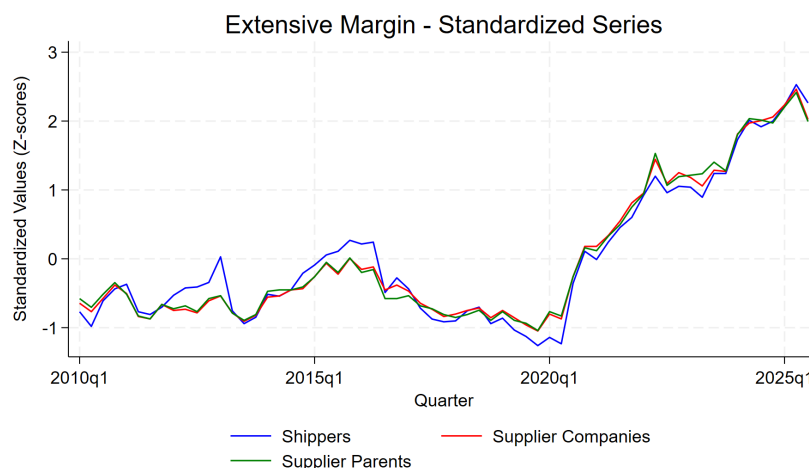


Figure 2: Ford's Supply Chain Adjustment(Extensive Margin) vs. Ford Delivery Risk due to port congestion

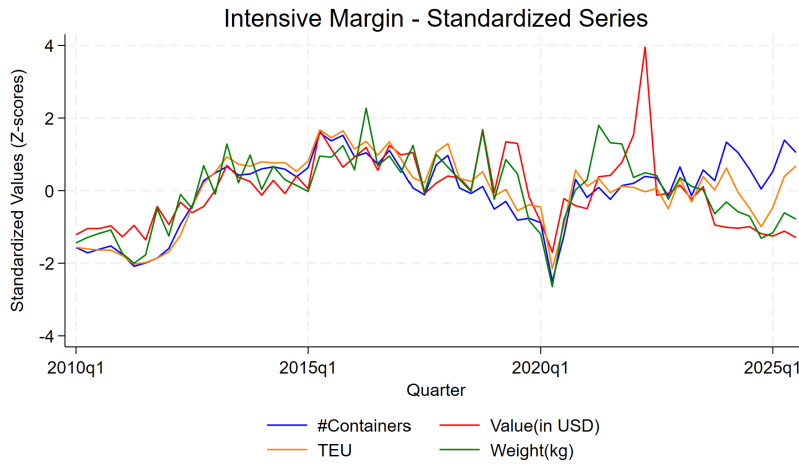


Figure 3: Ford's Supply Chain Adjustment(Intensive Margin) vs. Ford Delivery Risk due to port congestion

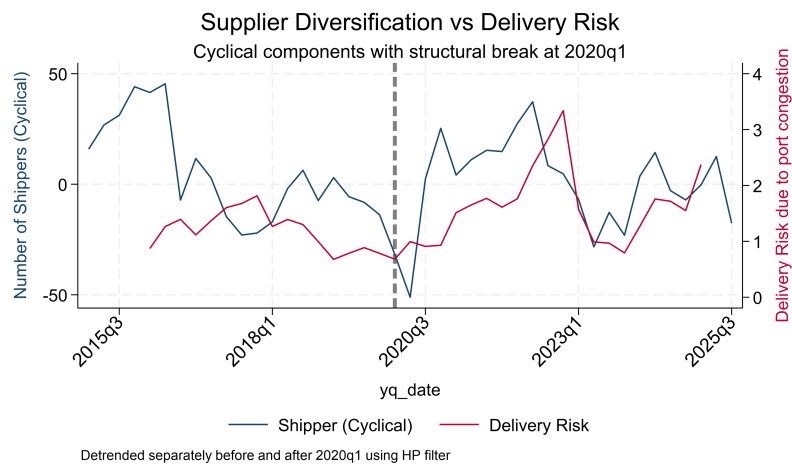


Figure 4: Ford's number of suppliers and Delivery risk due to Port Congestion

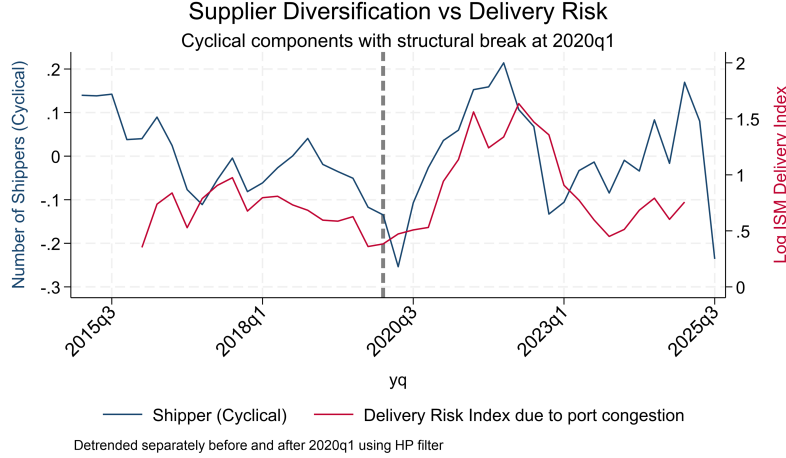


Figure 5: Average number of suppliers and Delivery risk due to Port Congestion across all firms

3.3 Supplier Sourcing Adjustment Upon Delivery risk

Armed with firm-level measures of delivery risk, this section documents reduced-form relationships between delivery risk and U.S. importers supplier-sourcing adjustments. Specifically, we estimate the following specification:

$$\ln(y_{i,t+h}) = \sum_{\ell=0}^3 \beta_{\ell} \ln(DeliveryRisk_{i,t-\ell}) + \mathbf{X}'_{i,t} \boldsymbol{\gamma} + \alpha_i + \delta_{s(i),t} + \varepsilon_{i,t+h} \quad (3)$$

where $y_{i,t+h}$ denotes firm i 's supplier-sourcing outcomes at horizon $t+h$, including the number of shipments, the number of suppliers, the number of HS2 product varieties, and the number of foreign sourcing countries. We currently set $h = 1$, corresponding to one-period-ahead sourcing adjustments. $DeliveryRisk_{i,t-\ell}$ is the firm-level delivery risk measure, constructed from the volatility of port congestion time and standardized to have mean zero and unit variance within the estimation sample. We include the contemporaneous value and up to three lags of delivery risk ($\ell = 3$) to allow for dynamic adjustment in firms sourcing decisions. $\mathbf{X}_{i,t}$ is a vector of firm-level controls, including firm sales, log total assets, market-to-book ratio, return on assets, and long-term book leverage. α_i denotes firm fixed effects, $\delta_{s(i),t}$ denotes NAICS 4-digit industry-by-time fixed effects, and $\varepsilon_{i,t+h}$ is an error term. The standard error is clustered at firm and time level.

The variable of interest is firm-level delivery risk. We examine how delivery risk affects firms forward-looking supplier-sourcing decisions, capturing both the contemporaneous response and the cumulative effect over time. Our shiftshare construction of delivery risk helps alleviate concerns about reverse causality, as it mitigates endogeneity arising from importers

endogenous choices of sourcing countries.

Table 3: Supplier Sourcing Adjustment upon Delivery Risk, Manufacturing

	(1) #Ship	(2) #Suppliers	(3) HS2 Varieties	(4) #Origin Countries
Delivery Risk	0.186*** (8.56)	0.123*** (7.97)	0.0722*** (7.06)	0.0804*** (8.09)
L.Delivery Risk	0.0323** (3.12)	0.0295* (2.73)	0.00516 (0.78)	0.00942 (1.31)
L2.Delivery Risk	0.0273* (2.61)	0.0252** (2.99)	0.00861 (1.19)	0.0174** (2.99)
L3.Delivery Risk	0.0274 (1.52)	0.0171 (1.45)	0.0206* (2.38)	0.0136 (1.71)
Cumulative (L0L3)	0.2732***	0.1953***	0.1067***	0.1207***
SE	0.0394	0.0253	0.0178	0.0169
N	18,324	18,324	18,324	18,324
Adj. R ²	0.843	0.844	0.792	0.854
Firm FE	yes	yes	yes	yes
NAICS4×Time FE	yes	yes	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.010$

Note: This table reports results using the firm-quarter dataset for the manufacturing sector over the period 2016-2025. Columns (1)-(4) examine the number of shipments, number of suppliers, number of HS2 product varieties, and number of origin countries, respectively. Delivery risk is measured as a firm-level exposure to the volatility of port congestion waiting times. All regressions control for log sales, log total assets, the book-to-market ratio, return on total assets, and long-term book leverage, and include firm fixed effects as well as 4-digit NAICS industry-by-time fixed effects. Standard errors are two-way clustered at the firm and yearquarter levels.

Table 3 presents the main findings using the volatility of port congestion time over the sample period 2016-2025. Our baseline analysis focuses on manufacturing firms, though we plan to extend the analysis to additional sectors, including construction, mining and quarrying, oil and gas, retail trade, utilities, and wholesale trade. Columns (1)-(4) document a positive and statistically significant relationship between delivery risk and firms supplier-sourcing adjustments. In Columns (1) and (2), we find that a one-standard-deviation increase in delivery risk - measured by the volatility of port congestion time - leads to an approximately 18.6 percent increase in the number of shipments and 12.3 percent increase in the number of suppliers in the subsequent period. These results indicate that firms actively diversify their sourcing activity in response to heightened delivery risk. The cumulative effect over four quarters is also significantly positive, implying that a one-standard-deviation in-

crease in delivery risk raises the number of shipments by 27.3 percent and the number of suppliers by 19.5 percent. We observe qualitatively similar effects for the number of HS2 product varieties and the number of origin countries, though the magnitudes are smaller. This difference likely reflects higher adjustment costs associated with diversifying product varieties and sourcing countries, as U.S. importers tend to rely more on increasing the number of shipments and suppliers as primary margins for mitigating delivery risk.

In addition to delivery risk, we also examine whether firms adjust their sourcing behavior in response to realized delivery delays themselves, measured by firms exposure to the level of port congestion time. To facilitate this comparison, we estimate the same empirical specification as in Equation (3), replacing the delivery risk variable with the firm-level delivery delay measure constructed as a weighted average of port congestion time levels across source countries.

Table 4 examines whether firms adjust their supplier sourcing behavior in response to delivery delays - the first moment of shipping conditions. Across all four sourcing margins - number of shipments, number of suppliers, HS2 varieties, and origin countries - we find no evidence that delivery delays induce systematic sourcing adjustments. The contemporaneous coefficients on delivery delay are small, negative, and statistically insignificant, and the same pattern holds for lagged effects up to three periods.

The negative sign of the coefficients is economically intuitive. Realized delivery delays mechanically imply that fewer shipments arrive within a given period, which can temporarily reduce observed shipment counts, supplier relationships, or sourcing variety. In this sense, the first-moment delivery delay captures an ex post realization of disruption rather than an ex ante risk that firms can actively hedge against. This mechanical effect contrasts sharply with the results in Table 3, where firms respond strongly and systematically to delivery risk, measured by the volatility of port congestion time. Together, the results suggest that it is not the level of delivery delays per se that drives sourcing adjustments, but rather the uncertainty surrounding delivery times.

Finally, we examine whether the level of delivery delays affects supplier-sourcing adjustments in response to delivery risk. To do so, we augment the baseline specification by including additional controls for firms exposure to realized delivery delays and its lags, measured as the weighted-average level of port congestion time. This allows us to assess whether controlling for average delivery conditions alters the estimated effect of delivery risk. Table 5 reports the results.

The results show that the precautionary sourcing response of U.S. importers to delivery risk remains strong and largely unchanged. Firms respond to heightened uncertainty in delivery conditions by diversifying their sourcing margins.

Table 4: Supplier Sourcing Adjustment upon Delivery Delay, Manufacturing

	(1) #Ship	(2) #Suppliers	(3) HS2 Varieties	(4) #Origin Countries
Delivery Time	-0.00554 (-0.22)	-0.00701 (-0.39)	-0.0196 (-1.75)	-0.00748 (-0.61)
L.Delivery Time	0.0191 (1.10)	0.00609 (0.50)	0.00711 (0.89)	0.00183 (0.21)
L2.Delivery Time	-0.00883 (-0.52)	0.000489 (0.04)	-0.00691 (-0.67)	-0.00258 (-0.28)
L3.Delivery Time	-0.0122 (-0.83)	-0.00637 (-0.55)	-0.000570 (-0.07)	-0.000164 (-0.02)
N	18,324	18,324	18,324	18,324
Adj. R ²	0.837	0.838	0.789	0.851
Firm FE	yes	yes	yes	yes
NAICS4×Time FE	yes	yes	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.010$

Note: This table reports results using the firmquarter dataset for the manufacturing sector over the period 2016:2025. Columns (1)-(4) examine the number of shipments, number of suppliers, number of HS2 product varieties, and number of origin countries, respectively. Delivery delay is measured as a firm-level exposure to the level of port congestion waiting times. All regressions control for log sales, log total assets, the book-to-market ratio, return on total assets, and long-term book leverage, and include firm fixed effects as well as 4-digit NAICS industry-by-time fixed effects. Standard errors are two-way clustered at the firm and yearquarter levels.

In contrast, the coefficients on delivery delay levels are negative and often statistically significant, suggesting that realized congestion and shipment delays reduce the volume of shipments and the number of suppliers that firms are able to receive ex post. Taken together, these results underscore that it is delivery risk, rather than realized delivery delays, that drives firms precautionary sourcing behavior. Firms adjust ex ante to increased uncertainty by diversifying their supplier base, even though ex post delays may depress observed sourcing activity.

A potential concern is whether the observed increase in supplier diversification reflects an ex-post response to realized delivery disruptions rather than a precautionary, forward-looking sourcing strategy. Two features of the data support an expectations-based interpretation. First, both the level and the volatility of port congestion are highly persistent over time. As shown in Table 6, country-level congestion volatility follows a strongly persistent process, with an autoregressive coefficient of approximately 0.7, implying that shocks to delivery risk decay only gradually. Consequently, current congestion volatility is informative about near-term future delivery conditions faced by importers sourcing from a given country.

Second, this persistence carries through to firms exposure to delivery risk. Firm-level delivery risk is constructed as a shift-share weighted average of country-level congestion volatility using pre-determined import shares. Consistent with the persistence of the underlying country-level process, firms delivery risk exposure is itself highly persistent over time, even after controlling for firm fixed effects and common time shocks (Table 6). Importantly, while both the level and the volatility of congestion are persistent, they capture distinct economic forces. Persistent congestion levels raise expected trade and working-capital costs, leading firms to contract sourcing activity along the intensive margin. In contrast, persistent volatility increases uncertainty about delivery outcomes, thereby raising the value of supplier diversification as a hedge against disruption risk. Together, these findings imply that a firms current delivery risk exposure provides a meaningful proxy for its expected near-term delivery risk, and that the positive relationship between delivery risk and subsequent supplier diversification reflects, at least in part, a precautionary ex-ante sourcing response to anticipated delivery risk rather than a purely ex-post reaction to realized disruptions.

Table 5: Supplier Sourcing Adjustment upon Delivery Risk, Manufacturing

	(1) #Ship	(2) #Suppliers	(3) HS2 Varieties	(4) #Origin Countries
Delivery Risk	0.223*** (9.84)	0.151*** (9.60)	0.0930*** (8.42)	0.0998*** (9.45)
L.Delivery Risk	0.0680*** (5.77)	0.0557*** (5.32)	0.0214** (2.89)	0.0264** (3.43)
L2.Delivery Risk	0.0519*** (4.54)	0.0417*** (4.78)	0.0182* (2.69)	0.0275*** (4.43)
L3.Delivery Risk	0.0349* (2.19)	0.0215 (1.97)	0.0217* (2.47)	0.0150* (2.08)
Delivery Time	-0.0482* (-2.12)	-0.0362* (-2.40)	-0.0368*** (-3.83)	-0.0263* (-2.36)
L.Delivery Time	-0.0423* (-2.12)	-0.0381* (-2.62)	-0.0168 (-1.87)	-0.0251* (-2.55)
L2.Delivery Time	-0.0689** (-3.56)	-0.0432* (-2.69)	-0.0305* (-2.62)	-0.0297** (-2.87)
L3.Delivery Time	-0.0437* (-2.41)	-0.0299* (-2.34)	-0.0143 (-1.59)	-0.0144 (-1.59)
N	18,324	18,324	18,324	18,324
Adj. R ²	0.846	0.847	0.794	0.856
Firm FE	yes	yes	yes	yes
NAICS4×Time FE	yes	yes	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.010$

Note: This table reports results using the firmquarter dataset for the manufacturing sector over the period 2016:2025. Columns (1)-(4) examine the number of shipments, number of suppliers, number of HS2 product varieties, and number of origin countries, respectively. Delivery risk is measured as a firm-level exposure to the volatility of port congestion waiting times. Delivery delays and their lags measured by firms weighted-average exposure to the level of port congestion time are also included as controls to account for the level of delivery delay. All regressions control for log sales, log total assets, the book-to-market ratio, return on total assets, and long-term book leverage, and include firm fixed effects as well as 4-digit NAICS industry-by-time fixed effects. Standard errors are two-way clustered at the firm and yearquarter levels.

Table 6: Persistence of Delivery Risk(Volatility) and Delivery Delays(Level)

	(1) Country Delivery Risk	(2) Country Delivery Delay	(3) Firm Delivery Risk	(4) Firm Delivery Delay
L.Country Delivery Risk	0.682*** (0.060)			
L.Country Delivery Time		0.638*** (0.048)		
L.Firm Delivery Risk			0.580*** (0.039)	
L.Firm Delivery Time				0.558*** (0.064)
Observations	864.000	864.000	141712.000	141712.000
Adj. ²	0.749	0.734	0.699	0.598

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports persistence regressions for delivery risk and delivery delays at the country and firm levels. Delivery risk is measured as a firm-level exposure to the volatility of port congestion waiting times. Delivery delays are measured by firms weighted-average exposure to the level of port congestion time. Columns (1) and (2) present country-level persistence estimates for delivery risk and delivery delay, respectively. Columns (3) and (4) present firm-level persistence estimates for shift-share delivery risk exposure and delivery delay exposure. Firm fixed effects and year-quarter fixed effects are controlled. Standard errors are two-way clustered at the firm and year-quarter levels.

3.4 Risk shifting upon Uncertainty

Previous section show US firm will seek multi-sourcing by increasing their number of suppliers when there is a delivery delay. Next we want to further investigate whether there is a risk-shifting behaviour in multi-sourcing when there is increasing risk, that US firms reduce reliance on suppliers with high risk and shift toward other suppliers. Since it is difficult to identify risk at the supplier level, we study the risk-shifting behavior upon risk by using the delivery risk at the country level, which is the volatility of waiting time due to port congestion in each country. We estimate the following distributed-lag specification at the firm–country–quarter level:

$$Y_{i,c,t} = \sum_{k=0}^3 \beta_k \text{DeliveryRisk}_{c,t-k} + \sum_{k=0}^3 \gamma_k \text{CongTime}_{c,t-k} + \mathbf{Z}'_{c,t} \boldsymbol{\pi} + \mathbf{X}'_{i,t} \boldsymbol{\theta} + \alpha_{i,c} + \delta_t + \lambda_{g(i),t} + \varepsilon_{i,c,t}, \quad (4)$$

where $Y_{i,c,t}$ is the outcome of interest for firm i and origin country c in quarter t (e.g., the number of suppliers from country c , the number of shipments from country c). $\text{DeliveryRisk}_{c,t}$ measures delivery risk in country c (e.g., the volatility of port congestion time), and $\text{CongTime}_{c,t}$ is the average level of port congestion time in country c . Both variables enter with up to three

quarterly lags to allow for gradual adjustments in sourcing relationships.

The vector $\mathbf{Z}_{c,t}$ contains foreign-country macroeconomic controls that may directly affect sourcing allocations across countries, including $\ln(\text{GDP}_{c,t})$, inflation, unemployment, and the policy rate. These controls capture time-varying changes in country-level economic conditions that may independently influence the attractiveness or capacity of suppliers in country c . The vector $\mathbf{X}_{i,t}$ includes time-varying firm controls (e.g., size, profitability, leverage, and inventory measures). The specification includes firm–country fixed effects $\alpha_{i,c}$, which absorb all time-invariant firm–country match characteristics (e.g., persistent sourcing relationships and baseline comparative advantage), year-quarter fixed effects δ_t , which control for common aggregate shocks, and NAICS4-by-time fixed effects $\lambda_{g(i),t}$, which flexibly control for sector-specific demand and cost shocks over time. Standard errors are two-way clustered at the firm and year-quarter levels.

The coefficients of interest are the β_k s, along with their cumulative effect obtained by summing the β_k terms. These estimates capture the relationship between delivery risk in a supplier country and U.S. importers allocation of their supply chains across countries.

Table 7 presents the result. When delivery risk increases in a given foreign country c , firms significantly reduce the number of suppliers sourced from that country. At the same time, they expand sourcing from suppliers located in other countries, leading to a positive and statistically significant increase in the total number of suppliers. As a result, the share of suppliers in country c both relative to suppliers in other countries and relative to total suppliers declines sharply. These patterns indicate that firms respond to country-specific delivery risk by actively reallocating their supplier base away from riskier locations and toward safer alternatives. Table 8 shows similar risk-shifting behavior in firms shipment patterns across countries. Taken together, the evidence from Tables 6 and 7 demonstrates that firms responses to delivery risk reflect not merely diversification across similar suppliers, but a systematic reallocation of sourcing toward lower-risk countries i.e., a risk-shifting behavior rather than pure diversification.

Table 7: Risk Shifting Behavior, Manufacturing

	(1)	(2)	(3)	(4)	(5)
	#Suppliers in C	#Suppliers not in C	#Total Suppliers	Share C vs not in C	Share C vs total
Delivery Risk in C	-0.0376* (-2.13)	0.0133 (1.40)	0.0116 (1.36)	-0.0502* (-2.69)	-0.0370* (-2.30)
L.Delivery Risk in C	0.0277 (1.98)	0.0196** (3.05)	0.0187** (3.46)	0.0123 (0.82)	0.0102 (0.75)
L2.Delivery Risk in C	-0.0306* (-2.41)	-0.00103 (-0.14)	-0.000641 (-0.10)	-0.0336* (-2.26)	-0.0286* (-2.06)
L3.Delivery Risk in C	-0.0166 (-1.37)	0.0126* (2.24)	0.00899 (1.77)	-0.0289* (-2.18)	-0.0248* (-2.06)
Cumulative Effect	-0.0572*	0.0445**	0.0386**	-0.1004***	-0.0802***
SE	0.0295	0.0179	0.0166	0.0320	0.0288
N	22,307	22,307	22,307	22,151	22,307
Adj. R ²	0.689	0.921	0.921	0.814	0.820
FirmCountry FE	yes	yes	yes	yes	yes
NAICS4×Time FE	yes	yes	yes	yes	yes

Notes: This regression is clustered at time and firm level

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table shows the result using the firm-supplier country-quarter dataset from 2016 to 2025.

Column(1) studies the number of suppliers in country C when delivery risk in country C increases. Column (2) studies the number of suppliers in countries other than C. Column(3) studies the total number of suppliers in all countries. Column (4) studies the fraction of suppliers in country C over suppliers not in country C. Column (5) studies the fraction of suppliers in country C over total suppliers; All regressions control for firm characteristics, including log total assets, book-to-market ratios, return on total assets, and long-term book leverage, as well as sourcing-country macroeconomic conditions such as GDP, CPI inflation, unemployment, and the policy interest rate. All specifications further include firm-supplier-country fixed effects, time fixed effects, and 4-digit NAICS-by-time fixed effects. Standard errors are two-way clustered at the firm and year-quarter levels.

Table 8: Risk Shifting Behavior, Manufacturing

	(1)	(2)	(3)	(4)	(5)
	#Ships from C	#Ships not from C	#Total Ships	Share C vs not in C	Share C vs total
Delivery Risk in C	-0.0488 (-1.57)	0.0207 (1.75)	0.0187 (1.97)	-0.0539* (-2.13)	-0.0469 (-2.02)
L.Delivery Risk in C	-0.00266 (-0.11)	0.0358*** (4.04)	0.0302*** (3.81)	-0.0359* (-2.22)	-0.0325 (-1.98)
L2.Delivery Risk in C	-0.0334 (-1.23)	0.00121 (0.13)	0.00337 (0.40)	-0.0230 (-1.23)	-0.0199 (-1.13)
L3.Congestion Vol in C	-0.0628* (-2.39)	0.0252** (3.05)	0.0164* (2.09)	-0.0706*** (-3.72)	-0.0633** (-3.53)
Cumulative (L0L3)	-0.1477**	0.0829***	0.0687***	-0.1835***	-0.1626***
SE	0.0571	0.0242	0.0223	0.0468	0.0434
N	22,307	22,307	22,307	22,155	22,307
Adj. R ²	0.736	0.922	0.920	0.801	0.793
FirmCountry FE	yes	yes	yes	yes	yes
NAICS4CETime FE	yes	yes	yes	yes	yes

Notes: This regression is clustered at time and firm level

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table shows the result using the firm-supplier country-quarter dataset from 2016 to 2025.

Column(1) studies the number of shipments in country C when delivery risk in country C increases. Column (2) studies the number of shipments in countries other than C. Column(3) studies the total number of shipments in all countries. Column (4) studies the fraction of shipments in country C over shipments not in country C. Column (5) studies the fraction of shipments in country C over total shipments; All regressions control for firm characteristics, including log total assets, book-to-market ratios, return on total assets, and long-term book leverage, as well as sourcing-country macroeconomic conditions such as GDP, CPI inflation, unemployment, and the policy interest rate. All specifications further include firm-supplier-country fixed effects, time fixed effects, and 4-digit NAICS-by-time fixed effects. Standard errors are two-way clustered at the firm and year-quarter levels.

3.5 Diversification and Firm Impact

Having established that delivery risk induces firms to diversify their supplier base, this section examines how such risk-induced diversification affects firm outcomes, focusing on the level and volatility of production costs and profit margins.

Shift-Share Approach A key identification challenge is that supplier diversification is an endogenous firm choice and may be correlated with unobserved firm characteristics - such as managerial quality, operational flexibility, or risk preferences that simultaneously influence sourcing decisions and firm performance. To establish a causal relationship, it is therefore necessary to isolate variation in diversification that is driven by exogenous delivery

risk rather than by firms underlying characteristics.

To address this concern, we adopt an instrumental variables strategy that exploits firm-level exposure to delivery risk. Specifically, we use the same shift-share measure of delivery risk constructed in the previous section - based on exogenous port congestion - induced waiting times - as an instrument for supplier diversification. This measure captures variation in diversification driven by congestion shocks at ports that are plausibly unrelated to firms contemporaneous production decisions, except through their impact on sourcing choices.

The validity of the instrument rests on two conditions. The relevance condition requires that the instrument strongly predicts supplier diversification. As shown in the previous section, firms with greater exposure to delivery risk significantly increase diversification, satisfying this requirement. The exclusion restriction requires that delivery risk affects firm outcome such as production costs and profit margins - only through its impact on diversification. This assumption is supported by several features of the empirical design. First, the port congestion measure is residualized to remove global trends, seasonality, and partner-country-specific components, limiting its direct correlation with firms cost and pricing conditions. Second, all specifications include firm fixed effects, time fixed effects, and a rich set of time-varying firm controls. After accounting for these factors, the remaining variation in delivery risk is plausibly exogenous and operates primarily through firms sourcing decisions, rather than directly affecting production costs or profit margins.

Two-Stage Least Squares Specification We estimate the causal effect of supplier diversification on firm outcomes using a two-stage least squares (2SLS) framework. The first stage quantifies how delivery risk induces supplier diversification and is specified as follows:

$$\ln(\text{Diversification}_{i,t}) = \alpha + \beta \ln(\text{DeliveryRisk}_{i,t}) + \gamma_i + \delta_t + \mathbf{X}'_{i,t}\boldsymbol{\theta} + \varepsilon_{i,t}, \quad (5)$$

where $\text{Diversification}_{i,t}$ denotes the number of suppliers firm i has in period t .

The second stage examines how delivery-risk-induced diversification affects firm outcomes, such as production costs, profit margins, and their corresponding volatilities. Using the predicted values of diversification from the first stage, we estimate:

$$\ln(Y_{i,t}) = \alpha + \beta \widehat{\ln(\text{Diversification})}_{i,t} + \gamma_i + \delta_t + \mathbf{X}'_{i,t}\boldsymbol{\theta} + u_{i,t}, \quad (6)$$

where $Y_{i,t}$ denotes the firm outcome of interest. All specifications include firm fixed effects γ_i , time fixed effects δ_t , and the same set of firm-level controls $\mathbf{X}_{i,t}$ as in the first stage.

In the current analysis, we focus on two firm outcome variables. Unit cost is defined as

the ratio of cost of goods sold to sales,

$$\text{UnitCost}_{i,t} = \frac{\text{COGS}_{i,t}}{\text{Sales}_{i,t}},$$

which captures average production costs relative to revenue. Profit margin is defined as the ratio of gross profits to sales,

$$\text{ProfitMargin}_{i,t} = \frac{\text{Sales}_{i,t} - \text{COGS}_{i,t}}{\text{Sales}_{i,t}}.$$

Both variables are constructed using Compustat data and are expressed in logarithms when used as dependent variables. One thing to highlight is that, the Cost of Goods Sold (COGS) encompasses all costs directly attributed to production, including materials, labor, and overhead cost, and is utilized to measure unit variable costs. COGS can be calculated using either the Last In, First Out (LIFO) or First In, First Out (FIFO) methods. To analyze how diversification affects production costs using the data, we restrict our regression to firms that employ the LIFO method, allowing changes in variable costs to be immediately reflected in the data.

The results of the two-stage least squares (2SLS) estimation are reported in Table 9. The first-stage results indicate that the instrument is a strong and statistically significant predictor of supplier diversification: the coefficient on delivery risk is positive and highly significant, confirming the relevance of the instrument.

Turning to the second-stage estimates, the coefficient on diversification is positive and statistically significant in the unit cost regression. Because the endogenous variable is standardized, the estimated coefficient implies that a one-standard-deviation increase in supplier diversification induced by delivery risk raises unit costs by approximately 1.1 percent. In contrast, the same increase in diversification leads to a statistically significant decline in profit margins of about 1.3 percent. These findings indicate that the precautionary sourcing strategies adopted by U.S. importers in response to delivery risk increase their production cost burden and compress profitability.

Table 9: IV Results: Supplier Diversification and Firm Outcomes

Panel A: Two-Stage Least Squares		
	(1)	(2)
	Unit Cost	Profit Margin
#Suppliers	0.0109*	-0.0133**
	(0.00571)	(0.00638)
Observations	8225	8225
Adj. ²	0.171	0.177
Panel B: First Stage		
	(1)	(2)
	#Suppliers	#Suppliers
Delivery Risk	0.177***	0.177***
	(0.0228)	(0.0228)
Observations	8225	8225
Adj. ²	0.854	0.854

Notes: Panel A reports 2SLS estimates of the effect of diversification on firm. Panel B reports the corresponding first stage. All specifications include firm and time fixed effects and the same controls.

Standard errors are clustered by firm and year-quarter.

Finally, we examine how delivery-risk-induced supplier diversification affects firm-level volatility, providing evidence in support of an efficiency-stability trade-off. Table 10 reports two-stage least squares estimates of the effect of delivery-risk-induced diversification on the volatility of production costs and profit margins.

Volatility is measured using a forward-looking window: for each firm-year, we compute the standard deviation of quarterly costs and profit margins over the current quarter and the subsequent three quarters. To ensure that the volatility measure is non-overlapping, we retain only observations from the first quarter of each calendar year, so that the regressions capture how diversification decisions made at the beginning of the year affect volatility over the remaining quarters of the same year. This construction yields a non-overlapping, annual measure of volatility and avoids the mechanical serial correlation inherent in rolling-window measures.

The results show that diversification significantly reduces volatility. A one-standard-deviation increase in supplier diversification lowers unit cost volatility by approximately 1.45 percent and reduces profit volatility by about 1 percent. These effects are statistically significant and economically meaningful.

Table 10: IV Results: Supplier Diversification and Firm Outcomes

Panel A: Two-Stage Least Squares		
	(1) Unit Cost Vol.	(2) Profit Vol.
#Suppliers	-0.0145* (0.00807)	-0.00944** (0.00401)
Observations	7093	7093
Adj. ²	0.0454	0.0404
Panel B: First Stage		
	(1) #Suppliers	(2) #Suppliers
Delivery Risk	0.166*** (0.0141)	0.166*** (0.0247)
Observations	7093	7093
Adj. ²	0.849	0.849

Notes: Panel A reports 2SLS estimates of the effect of diversification on firm. Panel B reports the corresponding first stage. All specifications include firm and year fixed effects and the same controls. Standard errors are clustered by firm and year.

Combined with earlier results showing that diversification raises production costs and compresses profit margins, these findings highlight a clear efficiency-stability trade-off: firms diversify their supplier base in response to delivery risk to smooth production and profits, but at the cost of higher average costs and lower profitability.

4 Model

In this section, we construct a model of an economy with endogenous firm sourcing under delivery risk, building on the framework proposed by Dhyne et al. 2023. Similar in *ibid.*, there exists a finite set of final goods producers, denoted by Ω , each producing a differentiated product. Final goods producers optimally choose their intermediate goods suppliers and form buyer-supplier linkages. Establishing these linkages is costly and entails a random, firm-pair-specific fixed cost. We assume that the final goods producer (the buyer) initiates the relationship and bears the associated fixed cost.

The key departure from the literature is the introduction of delivery risk caused by potential supply chain disruption, which affects the fraction of an order that intermediate goods suppliers can fulfill. Importantly, final goods producers must choose their set of suppliers and incur the relationship-specific fixed costs before the delivery shock is realized. Once the delivery shock occurs, the final goods producers decide on production and place orders only

with the subset of suppliers with whom they have pre-established relationships.

Additionally, we assume final goods producers are risk averse, meaning they consider not only expected profits but also the volatility of profits under a given sourcing strategy. As a result, firms may diversify their supplier base and shift away from the most cost-effective suppliers toward more stable ones, thereby mitigating supply chain risk. We currently focus on a closed economy but will consider extend the model to an open economy with export import in the future.

For notation, in the following part, whenever a variable has two subscripts ij , the first refers to the origin of the good and the second to its destination.

4.1 Preferences and Demand

We assume the consumers in the economy supplies L unit of labor inelastically to all the firms and have identical, homothetic CES preferences over differentiated non-traded consumption goods produced by final good producer with elasticity of substitution $\sigma > 1$:

$$U = \left(\sum_{k \in \Omega} (\beta_{jH} q_{jH})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (7)$$

where Ω denotes the set of final goods in the economy, j denotes a final good producer that sell goods to household, and H denotes the domestic final demand from households. The term β_{jH} captures how salient the good produced by firm j is in the final consumption bundle.

The budget constraint of the household is

$$E = \sum_{j \in \Omega} p_{jH} q_{jH} \quad (8)$$

where E denotes the aggregate expenditure.

The optimization problem of the household is

$$\max \left(\sum_{k \in \Omega} (\beta_{jH} q_{jH})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \text{ st. } E = \sum_{j \in \Omega} p_{jH} q_{jH} \quad (9)$$

Standard algebra could derive following demand function that

$$q_{jH} = \beta_{jH}^{\sigma-1} \frac{p_{jH}^{-\sigma}}{P^{1-\sigma}} E \quad (10)$$

where P denotes the domestic consumer price index:

$$P = \left(\sum_{k \in \Omega} \beta_{jH}^{\sigma-1} p_{jH}^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \quad (11)$$

Detail algebra for demand function and price index see appendix B. We assume that final goods are substitutes, and therefore $\sigma > 1$.

4.2 Timing of the model

The model currently is featured as a two-stages static model.

- **Ex-ante sourcing:** In stage 1, each finalgoods producer forms expectations about both aggregate and supplierspecific delivery risks. Based on these beliefs, the firm chooses its set of intermedategoods suppliers and pays the relationship-specific fixed costs to establish those linkages.
- **Delivery realization and production:** In stage 2, delivery shocks materialize: each supplier either fulfills the entire order or delivers only a fraction, impacting suppliers' productivity or pricing of input. Given these realized shocks, the final good producer orders inputs and hires laborbut it may only source from those suppliers with whom it formed a relationship in stage 1

4.3 Intermediate good supplier

Intermediate goods suppliers i use only labor as an input and sell their output to final goods producers. Their production technology is linear:

$$q_i = z_i \ell_i,$$

where z_i is a stochastic, idiosyncratic productivity term drawn independently from a log-normal distribution, consistent with the framework in Dhyne et al. 2023. This distribution applies to both intermediate and final goods producers. The marginal cost for supplier i is given by $\frac{w}{z_i}$, where w is the wage rate.

Let p_{ij} denote the price charged by supplier i to buyer j . We assume that supplier i sets this price as a constant markup over marginal cost:

$$p_{ij} = \mu_{ij} \cdot \frac{w}{z_i},$$

where μ_{ij} is an exogenous, pair-specific markup. In the baseline model, we set $\mu_{ij} = 1$, implying zero profits from supplier-buyer relationships and simplifying the analysis of sourcing decisions.

Intermediate suppliers face a common aggregate delivery shock, which occurs with probability ρ_{agg} . When this shock materializes, every supplier i delivers only a fraction $\Gamma_i < 1$ of the order placed by its buyer; In the absence of the shock it delivers fully (i.e. $\Gamma_i = 1$). We assume that the post-shock delivery fraction Γ_i is idiosyncratic and negatively related to the suppliers productivity. Specifically, if z_i denotes supplier i 's productivity, and $\tilde{z}_i \in [0, 1]$ is its normalization, we have

$$\Gamma_i = \begin{cases} a - b\tilde{z}_i, & \text{with probability } \rho_{\text{agg}}, \\ 1, & \text{with probability } 1 - \rho_{\text{agg}}, \end{cases} \quad \text{with } a, b > 0,$$

so that more productive suppliers (higher $\tilde{z}_i$) face smaller Γ_i and hence larger disruptions when the aggregate shock occurs. This generates a trade-off in the sourcing decision: high-productivity suppliers offer lower prices but come with greater disruption when shock occur. As a result, final goods producers may prefer to source from less productive but more reliable suppliers.

While our model currently focuses on a closed economy, this assumption mirrors real-world trade-offs in global supply chains. For instance, suppliers in China or energy exporters like Russia often offer cost-effective, high-productivity inputs. However, these inputs may come with elevated geopolitical or logistical risks, such as tariffs, embargoes, or transportation bottlenecks. Consequently, firms in North America or Europe may choose to diversify away from such suppliers in favor of more stable alternatives, even at the cost of lower efficiency.

4.4 Final good producers

Each final goods producer j produces a unique, differentiated variety. Production relies on both labor and a set of intermediate inputs sourced from a subset of eligible intermediate goods suppliers. Final goods are then sold to a representative final consumer.

The producer takes as given the supplier-specific input prices p_{ij} , the wage rate w , the aggregate expenditure E , and the consumer price index P . The production function of firm j is given by a CES aggregator of labor and intermediate inputs:

$$q_j = \phi_j \left(\sum_{i \in Z_j} (\alpha_{ij} m_{ij})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj} \ell_j)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}},$$

where $Z_j \subseteq \mathcal{Z}_j$ denotes the set of suppliers from which firm j has formed ex-ante relationships and is eligible to source inputs; ϕ_j is the idiosyncratic productivity (TFP) of firm j ; α_{ij} captures the relative importance of input i for firm j ; α_{Lj} captures the weight of labor ℓ_j in the production process; m_{ij} is the effective quantity of intermediate input i received by firm j .

Intermediate inputs are subject to delivery risk. Let q_{ij} be the quantity of input ordered by firm j from supplier i , and let $\Gamma_{ij} \in (0, 1)$ denote the realized delivery shock, with $\Gamma_{ij} < 1$ representing partial delivery. Then the quantity of input effectively received is:

$$m_{ij} = \Gamma_{ij} q_{ij}.$$

We assume that delivery shocks are destructive — that is, the undelivered portion of each order is lost and that final goods producers optimally utilize all inputs that are actually delivered. Hence, the identity $m_{ij} = \Gamma_{ij} q_{ij}$ holds with equality.

Let $\Delta_j = (\Gamma_{1j}, \Gamma_{2j}, \dots, \Gamma_{Ij})^\top$ denote the vector of realized delivery outcomes from all potential suppliers. The decision making process of final good producer is that

1. **Ex-ante (before realization of delivery shocks Δ_j):** The firm chooses its sourcing strategy $Z_j \subseteq \mathcal{Z}_j$, i.e., the set of suppliers with which to form fixed-cost linkages.
2. **Ex-post (after realization of Δ_j):** The firm chooses input orders $\{q_{ij}\}_{i \in Z_j}$ and labor ℓ_j , taking into account the realized availability of intermediate inputs.

4.4.1 Unit Cost

The cost minimization problem will yield the unit cost of production, given the shock is realized

$$\min \sum_{i \in Z_j} p_{ij} q_{ij} + w \ell_j$$

$$\text{st. } q_j = \phi_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij} q_{ij})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj} \ell_j)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

FOC to production input and some algebra, I derive the analytical solution for unit cost function:

$$c_j = \frac{C_j}{q_j} = \frac{1}{\phi_j} \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma} \right)^{\frac{1}{1-\sigma}} = \frac{1}{\phi_j} \Theta_j^{\frac{1}{1-\sigma}}$$

where term Θ is defined as the of firm j in the form that $\Theta_j = \sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} +$

$$\alpha_{Lj}^{\sigma-1} w^{1-\sigma}$$

Detail algebra please refer Appendix C

4.4.2 Sales to Household

In the preference and demand section (in section 1.1), we have derive that $q_{jH} = \beta_{jH}^{\sigma-1} \frac{p_{jH}^{-\sigma}}{P^{1-\sigma}} E$
So final goods producer j 's sales to household x_{jH} is

$$x_{jH} = p_{jH} q_j = \beta_{jH}^{\sigma-1} P^{\sigma-1} (p_{jH})^{1-\sigma} E$$

We assume that a firm sets a price that is equal to an exogenous markup times its marginal cost, so that

$$p_{jH} = c_j \mu_{jH}, P = \left(\sum_{j \in \Omega} \beta_{jH}^{\sigma-1} p_{jH}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$$

Since we derive $c_j = \frac{1}{\phi_j} \Theta_j^{\frac{1}{1-\sigma}}$ Therefore,

$$p_{jH} = c_j \mu_{jH} = \frac{1}{\phi_j} \Theta_j^{\frac{1}{1-\sigma}} \mu_{jH}$$

Substitute p_{jH} into x_{jH} , we have

$$x_{jH} = \beta_{jH}^{\sigma-1} P^{\sigma-1} p_{jH}^{1-\sigma} E = \beta_{jH}^{\sigma-1} \mu_{jH}^{1-\sigma} \phi_j^{\sigma-1} \Theta_j \frac{E}{P^{1-\sigma}}$$

4.4.3 Producer's cost shares

In Appendix C, we have derived that

$$q_{ij} = \frac{C_j}{\Theta_j} \cdot \frac{(\alpha_{ij} \Gamma_{ij})^{\sigma-1}}{p_{ij}^{\sigma}}$$

and

$$l_j = \frac{C_j}{\Theta_j} \cdot \frac{(\alpha_{Lj})^{\sigma-1}}{w^{\sigma}}$$

Therefore, the share of variable costs by final good producer j that is spend on intermediate inputs produced by firm $i \in Z_j$ is

$$s_{ij} = \frac{p_{ij} q_{ij}}{c_j q_j} = \frac{p_{ij} q_{ij}}{C_j} = \frac{(\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma}}{\Theta_j}$$

Analogously, the share of variable costs by firm j that is spend on labor is

$$s_{Lj} = \frac{w\ell_j}{c_j q_j} = \frac{\alpha_{Lj}^{\sigma-1} w^{1-\sigma}}{\Theta_j}$$

4.4.4 Producer's Sales and Profit

Given a sourcing strategy Z_j and a realization of delivery risk Δ , the variable profit of the final goods producer is

$$\pi_{jH} = x_{jH} - c_j q_{jH}$$

Since $c_j = \frac{1}{\phi_j} \Theta_j^{\frac{1}{1-\sigma}}$ and $q_{jH} = \beta_{jH}^{\sigma-1} \frac{p_{jH}^{-\sigma}}{p^{1-\sigma}} E = \beta_{jH}^{\sigma-1} \frac{(\mu_{jH} c_j)^{-\sigma}}{p^{1-\sigma}} E$ we have

$$c_j q_{jH} = \beta_{jH}^{\sigma-1} \mu_{jH}^{-\sigma} c_j^{1-\sigma} \frac{E}{p^{1-\sigma}} = \beta_{jH}^{\sigma-1} \mu_{jH}^{-\sigma} \phi_j^{\sigma-1} \Theta_j \frac{E}{p^{1-\sigma}}$$

Therefore, final goods producer's variable profit is

$$\pi_j^v = x_{jH} - c_j q_{jH} = \beta_{jH}^{\sigma-1} \mu_{jH}^{1-\sigma} \phi_j^{\sigma-1} \Theta_j \frac{E}{p^{1-\sigma}} - \beta_{jH}^{\sigma-1} \mu_{jH}^{-\sigma} \phi_j^{\sigma-1} \Theta_j \frac{E}{p^{1-\sigma}} \quad (12)$$

$$= \left(1 - \frac{1}{\mu_{jH}}\right) \beta_{jH}^{\sigma-1} \mu_{jH}^{1-\sigma} \phi_j^{\sigma-1} \Theta_j \frac{E}{p^{1-\sigma}} \quad (13)$$

$$= \Psi_j \Theta_j \frac{E}{p^{1-\sigma}} \quad (14)$$

where $\Psi_j = \left(1 - \frac{1}{\mu_{jH}}\right) \beta_{jH}^{\sigma-1} \mu_{jH}^{1-\sigma} \phi_j^{\sigma-1}$, $\Theta_j = \sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma}$. The supply chain shock is included in Θ_j .

The profits of final goods producer j are equal to variable profits less the fixed cost of link formation $\sum_{j \in Z_j} f_{ij} w$:

$$\pi_j(Z_j) = \Psi_j \Theta_j(Z_j) \frac{E}{p^{1-\sigma}} - \sum_{i \in Z_j} f_{ij} w. \quad (15)$$

Forming linkages is costly, and we assume that the final goods producer initiates the link formation and pays a random, firm-pair-specific fixed cost, denoted by f_{ij} . In this setup, the final goods producer charges a monopolistically competitive markup to households ($\mu_{jH} = \frac{\sigma}{\sigma-1}$) but possesses full bargaining power over suppliers ($\mu_{ij} = 1$), such that each supplier sells to the producer at marginal cost. Final good producer take the household demand shifter $\frac{E}{p^{1-\sigma}}$ as given when optimize.

4.4.5 Optimization of Risk Aversion Producer

The final goods producer faces an optimization problem in which, prior to the realization of shocks, it chooses a sourcing strategy Z_j to maximize expected utility. We assume that firm j is exogenously matched with a set of potential suppliers $\{Z_{ij}\}$, from which it endogenously selects a subset through optimization.

If firms are assumed ***risk-neutral*** typically, the objective is to maximizing expected profits. As a result, firms always select the lowest-cost set of suppliers, with no incentive to diversify. This leads to solutions in which firms concentrate sourcing on a single most cost-effective partner.

To generate meaningful sourcing heterogeneity and capture the empirical relevance of supply chain risk, we introduce ***risk aversion*** into the firms decision-making. Specifically, we assume that firms incur disutility from profit volatility, captured through a mean - variance utility formulation:

$$\max_{Z_j} \mathbb{E}_\Delta [\pi_j(Z_j, \Delta)] - \lambda \cdot \sqrt{\text{Var}_\Delta (\pi_j(Z_j, \Delta))},$$

where $\pi_j(Z_j, \Delta)$ is the firms profit under supply shock Δ , and λ is a risk-aversion parameter. This approach to modeling risk aversionsimple yet tractable follows a literature including gervais2018uncertainty, esposito2022demand, and esposito2024input.

Once the sourcing strategy is set ex ante, ex post profits are realized depending on the shock realization. The key tradeoff faced by the risk-averse firm is between cost efficiency, supply volatility, and fixed costs. Adding a supplier may reduce expected variable costs or diversify risk, thereby increasing expected profits or reducing profit variance, both of which raise the firm's overall utility. However, it also entails an additional fixed cost of forming the relationship. Importantly, because the volatility of variable profits depends on the covariance structure across suppliers, the marginal value of adding a supplier depends on the existing portfolio. That is, the gains from adding additional suppliers are *non-additive* and *interdependent*, making the decision combinatorially complex. Firms must weigh the incremental benefit of improved utility against rising costs in a networked, state-contingent environment.

4.5 Equilibrium

We consider an equilibrium where firms maximize expected utility, the household maximize consumption utility and goods market clear, labor market clear.

We assume that the firm's profits are distributed to workers. Hense, aggregate household

expenditure is given by

$$E = wL + \sum_{j \in \Omega} \pi_j$$

Labor market clearing implies that labor income is equal to firm's labor costs.

$$wL = \sum_{j \in \Omega} \frac{1}{\mu_j} s_{Lj} x_j$$

An equilibrium is characterized by the condition that all economic agents - final goods firms, intermediate goods firms, and the representative households simultaneously solve their respective optimization problems, taking prices, wages, and other parameters as given. Market clearing conditions in both goods and labor markets jointly determine the equilibrium values of aggregate variables.

Currently, we determine the general equilibrium by finding the fixed point of the aggregate demand shifter, $\frac{E}{p^{1-\sigma}}$, and the wage w in both aggregate states - when an aggregate delay occurs and when it does not.

4.6 Two suppliers example

Consider a simple case with only one final goods producer and two intermediate good suppliers. Supplier 1 is risky and receive a shock $\Gamma_1 < 1$ with probability ρ_1 , and supplier 2 is a safe supplier. So that

$$E_{\Gamma_1} = \rho_1 \Gamma_1 + (1 - \rho_1), E_{\Gamma_2} = 1$$

We also assume that the productivity of supplier 1 is higher than supplier 2 otherwise final goods producer will unambiguously choose supplier 2.

The expected profit for final goods producer is

$$\begin{aligned} \mathbb{E}\Pi_j = & \rho_1 \Psi_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma} \right) \frac{E}{p^{1-\sigma}} \\ & + (1 - \rho_1) \Psi_j \left(\sum_{i \in Z_j} (\alpha_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma} \right) \frac{E}{p^{1-\sigma}} \\ & - \sum_{i \in Z_j} f_{ij} w. \end{aligned} \quad (16)$$

Below are the simulation results of the supplier sourcing model. The horizontal axis represents the probability of delivery risk for supplier 1, ranging from 0 to 1.

Figure 6 illustrates the supplier sourcing strategy of the final goods firm. The figure shows that when the risk associated with supplier 1 is low, the firm will only source from

supplier 1, the more productive and cost-efficient supplier. As the risk increases, the firm begins to incorporate a second supplier, which is safer but more expensive, leading to supplier diversification. If the risk associated with supplier 1 continues to rise, the firm will eventually shift entirely to sourcing from supplier 2.

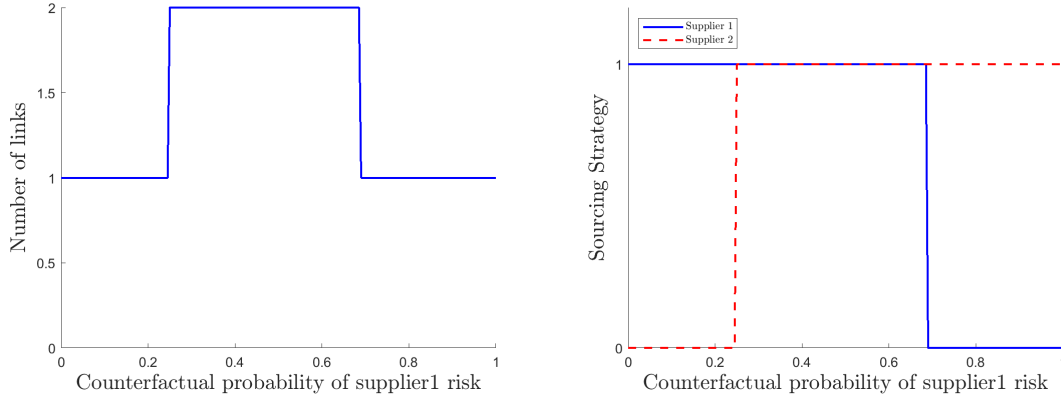


Figure 6: Supplier Sourcing

Figure 7 illustrates the unit costs, including variable cost, fixed cost, and total cost, as the delivery risk of supplier 1 increases. For the unit variable cost, the figure shows that sourcing solely from supplier 1 results in lower costs. However, the cost increases sharply when diversification begins and becomes even higher when sourcing exclusively from safe but less productive suppliers. Fixed costs per unit rises during the diversification phase. Similarly, the unit total cost figure indicates that the total cost during the diversification phase is higher than the initial cost when there is no risk. Thus, the model generates the result that diversification leads to higher costs.

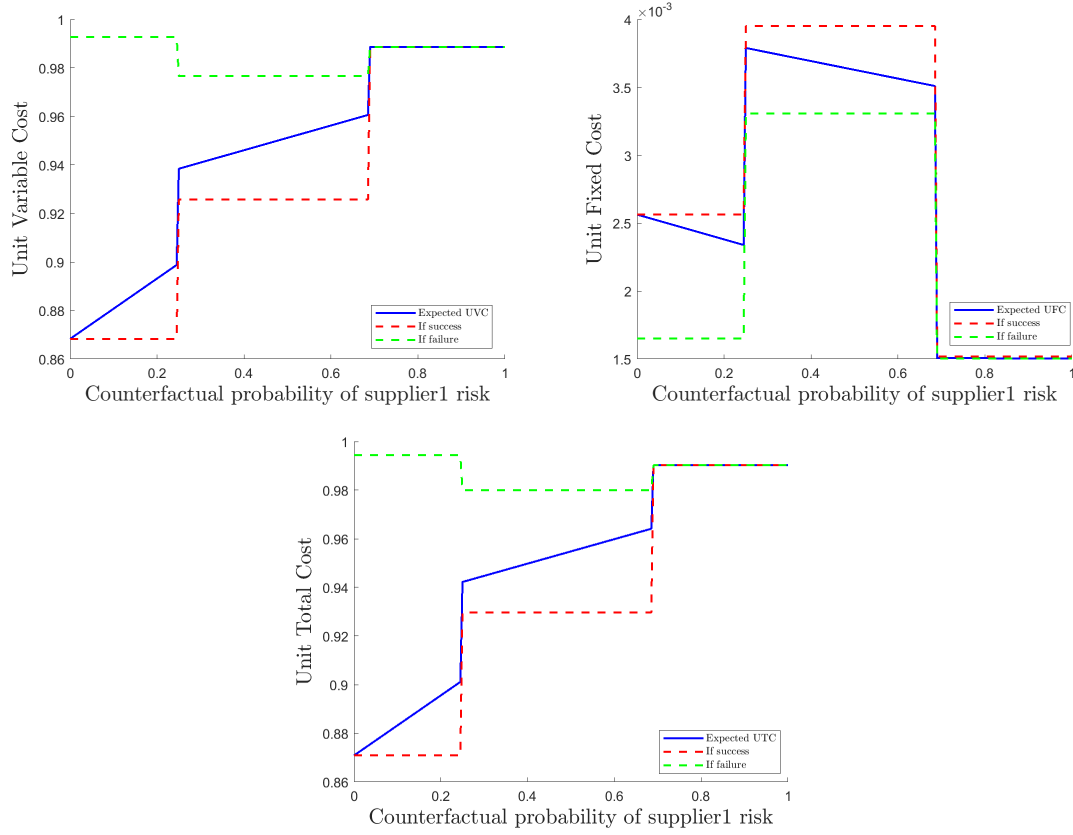


Figure 7: Unit Cost

Figure 8 illustrates the cost volatility as the delivery risk increases. The figure shows that volatility is high when sourcing exclusively from supplier 1, decreases sharply during the diversification phase, and drops to zero when sourcing exclusively from supplier 2. Thus, the model successfully captures the trade-off between cost level and cost volatility faced by producers under risk.

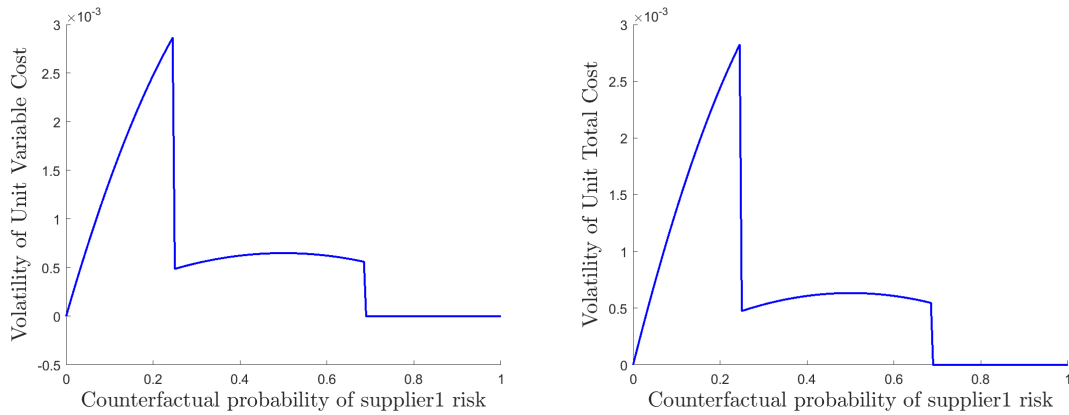


Figure 8: Cost Volatility

Figure 9 illustrates the expected utility under different sourcing strategies, defined as the expected profit minus the volatility of profit. The figure shows that as the risk increases, the profit becomes lower during the diversification phase, while the volatility also decreases. This highlights the importance of the risk-averse assumption. Without this assumption, a risk-neutral producer would always choose the supplier that maximizes profit, avoiding diversification altogether.

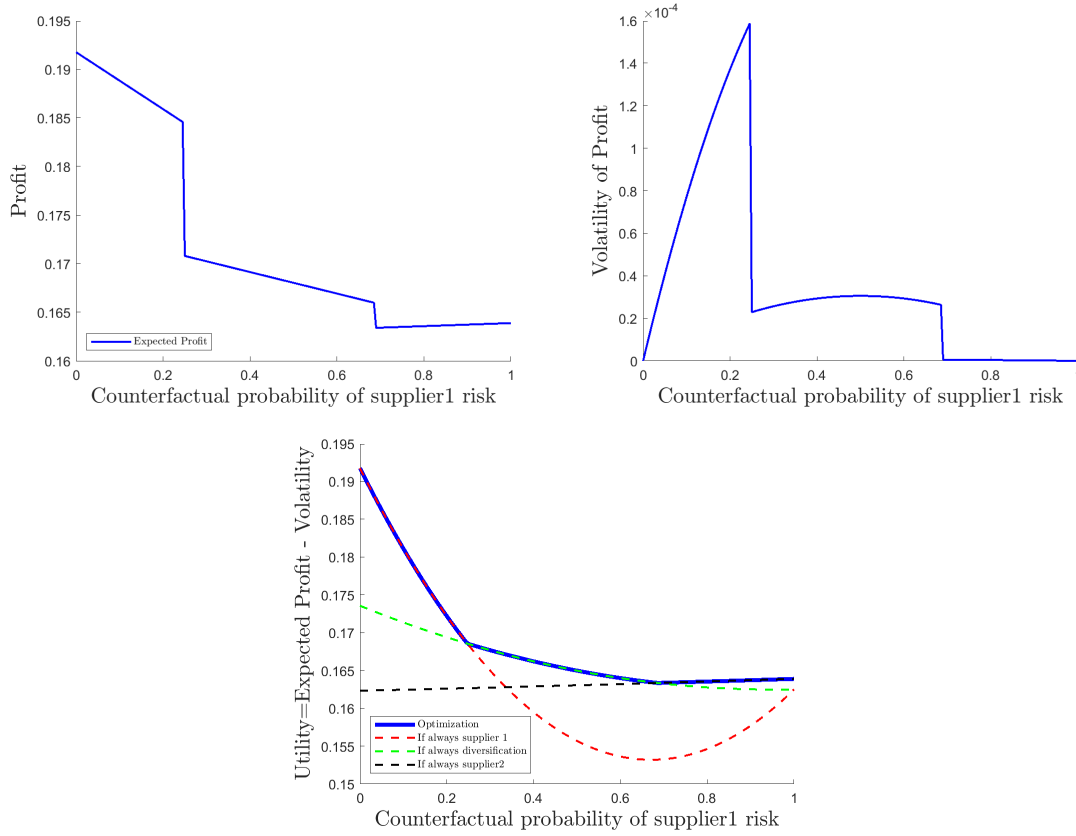


Figure 9: Utility

Figure 10 presents the aggregate results in equilibrium. It shows that the aggregate price level increases during the diversification phase, while aggregate income declines. This has significant economic implications: In a world with delivery uncertainty, private agents have incentives to mitigate risk through diversification strategies. While diversification strategy stabilize profits, they also increase costs, thereby driving up aggregate prices and reduce aggregate income. As a result, we are likely to continue living in a world with elevated price and subdued real income, even if shock doesn't realize.

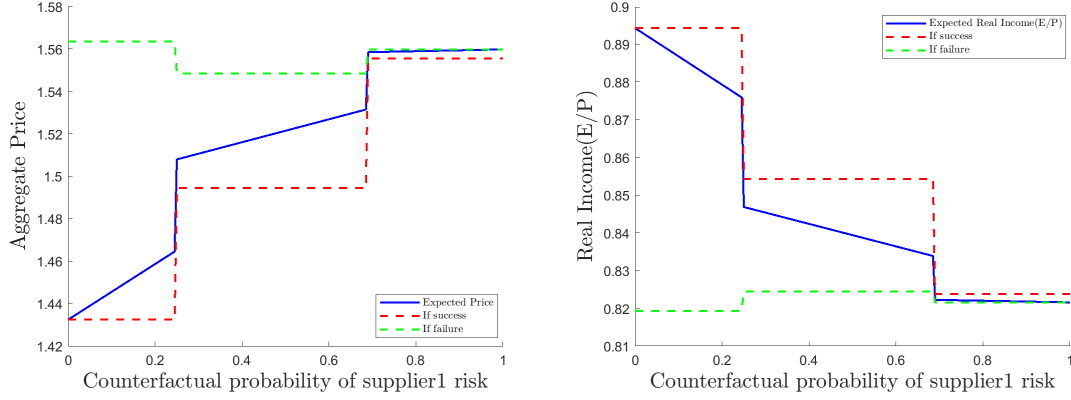


Figure 10: Aggregate Outcome

5 Quantitative Analysis

In this section, we generalize our model to an economy with a large number of intermediate suppliers and final-good producers. In the quantitative framework, a final-good producer is characterized by a set of potential suppliers, a productivity level, a vector of firmpair-specific weights, a vector of fixed cost draws, a vector of input cost draws, and a vector of delivery delay probabilities for all potential suppliers. The productivity levels of all firms (both intermediate suppliers and final-good producers), the firmpair-specific weights, as well as the fixed costs of establishing relationships, are stochastic and drawn from underlying distributions.

5.1 Calibration

- **Parameters from the literature**

We begin by taking several parameter values from existing literature with broad consensus. We set the demand elasticity (σ) to 4, following common practice in the literature, including Antras, Fort, and Tintelnot 2017 and Dhyne et al. 2023. We normalize firm-specific labor productivity shifters to $\alpha_{Lj} = 1$, domestic final demand shifters to $\beta_{jH} = 1$, and normalize the sum of firm-specific input shifters $\sum_{i \in Z_j} \alpha_{ij}^{\frac{\sigma-1}{\sigma}}$ to 1, so that the aggregate weight of intermediate inputs is equal to that of labor in the production function.

- **Parameterization of productivity draws, fixed cost draws, and input weight draws**

We assume that the productivity of both final goods producers and intermediate suppliers is drawn independently from a log-normal distribution, parameterized by a scale Φ_{scale}^ϕ and a dispersion Φ_{disp}^ϕ to be estimated.

We also assume the firm-pair-specific random variables, α_{ij} are independent draws from a log-normal distribution with scale parameters Φ_{scale}^α and dispersion parameters Φ_{disp}^α

Fixed costs for supplier relationships are also drawn independently from a log-normal distribution with scale and dispersion parameters Φ_{scale}^{fc} and Φ_{disp}^{fc} , respectively.

A key parameter in our model is the firm's risk aversion λ , which governs the utility penalty from profit volatility. This parameter can be disciplined using estimates of the elasticity of the number of suppliers with respect to delivery risk.

In principle, the full parameter set could be estimated via a method of simulated moments, following Dhyne et al. 2023, by targeting empirical moments such as the 50th, 75th, and 95th percentiles of the number of suppliers distribution, the labor cost share, firm-to-firm input shares, and the elasticity of the extensive margin of supplier sourcing with respect to delivery risk, as well as the correlation between input cost and delivery risk.

However, at the current stage, we temporarily adopt some parameter values from *ibid.*, estimated using Belgian firm-level transaction data: $\Phi_{scale}^\phi = -0.68$, $\Phi_{disp}^\phi = 0.56$, $\Phi_{scale}^\alpha = -3.2$, $\Phi_{disp}^\alpha = 0.93$, $\Phi_{scale}^{fc} = -5.71$, and $\Phi_{disp}^{fc} = 3.05$.

• Parameterization of delay probabilities and input disruption severity

We assume that suppliers are subject to a common aggregate delivery delay shock, which occurs with probability $\rho_{agg} \in (0, 1)$. When this shock materializes, *all* suppliers are affected. However, the *severity* of the disruption, denoted by Γ_i , varies across suppliers depending on their productivity.

Specifically, the disruption Γ_i is assumed to be negatively correlated with suppliers' productivity z_i , capturing a trade-off between efficiency and resilience: more productive suppliers tend to face greater vulnerability to delivery disruptions. We model the disruption intensity as:

$$\Gamma_i = a - b\tilde{z}_i, \quad \text{with } a, b > 0,$$

where $\tilde{z}_i \in (0, 1)$ is the normalized productivity of supplier i . Thus, more productive suppliers (higher $\tilde{z}_i$) experience more severe disruptions (lower Γ_i) when the aggregate shock occurs. Importantly, in the absence of the aggregate shock, no delivery disruption occurs and $\Gamma_i = 1$.

Pre-pandemic research by the McKinsey Global Institute estimates that, on average, firms experience a major supply chain disruption lasting one to two months every 3.7 years. This corresponds to an annual probability of approximately $1/3.7 \approx 27\%$.[§] Based on this evidence, we set the baseline value of the aggregate delivery disruption probability to $\rho_{agg} = 0.3$.

[§]McKinsey Global Institute (2020), *Future-proofing the supply chain*, <https://www.mckinsey.com/capabilities/operations/our-insights/future-proofing-the-supply-chain>

In our calibration, we currently set $a = 1$ and $b = 0.99$, which results in an average delivery disruption severity of approximately 22%, corresponding to a mean value of $\Gamma_i = 0.78$.

• Parameterization of Risk Aversion Parameter

For the risk aversion parameter, we calibrate $\lambda = 1.5$.[¶] This calibration allows us to match the elasticity of the number of suppliers with respect to the delivery delay index observed in the data.

We need to translate our theoretical delivery risk parameter ρ_{agg} into an empirically observable measure, so we map values of the aggregate component ρ_{agg} to the ISM Supplier Deliveries Index using a piecewise linear function. Specifically, we set $\rho_{agg} = 0.5$ to correspond to an ISM value of 50, representing normal delivery times. The mapping spans ISM values from 30 (faster deliveries, $\rho_{agg} = 0$) to 80 (severe delays, $\rho_{agg} = 1.0$), such that higher ρ_{agg} values represent progressively worse delivery performance as evaluated by industry practitioners.

We focus on the range where ρ_{agg} varies from 0.3 to 0.6, corresponding to ISM index values between 50 and 65 during the 2010–2020 period used in our regression analysis. With risk aversion parameter $\lambda = 1.5$, the elasticity of the number of suppliers with respect to ρ_{agg} within the range from 0.3 to 0.6 is 0.327. The corresponding elasticity with respect to the delivery delay index within the range of 50–65 is 1.051, which closely matches the estimated elasticity of 1.3 found in the regression.

Table 11: Model Parameters and Calibration

Symbol	Description	Value	Source / Note
σ	Elasticity of substitution	4	Antras, Fort, and Tintelnot 2017; Dhyne et al. 2023
Φ_{scale}^ϕ	Productivity scale (log-normal)	−0.68	Dhyne et al. 2023
Φ_{disp}^ϕ	Productivity dispersion	0.56	Dhyne et al. 2023
Φ_{scale}^α	Match quality scale	−3.2	Dhyne et al. 2023
Φ_{disp}^α	Match quality dispersion	0.93	Dhyne et al. 2023
Φ_{scale}^{fc}	Fixed cost scale	−5.71	Dhyne et al. 2023
Φ_{disp}^{fc}	Fixed cost dispersion	3.05	Dhyne et al. 2023
λ	Risk aversion parameter	1.5	match diversification elasticity
a, b	Idiosyncratic Deliver Disruption ($\Gamma_i = a + b\bar{z}_i$)	$a = 1, b = 0.99$	temporarily value

[¶]This value is lower than the estimate of 4 reported in Blaum, Esposito, and Heise 2025. However, their specification penalizes the variance of profits, whereas we penalize the standard deviation, which differs in scale.

5.2 Quantitative Results

[¶] Below are the counterfactual results showing how the firm's supplier sourcing strategy responds to changes in aggregate delivery risk, and how this in turn affects aggregate real income and aggregate prices.

Figure 11 illustrates how the average number of supplier links varies with increasing aggregate delay probability. When final good producers are risk-averse, higher delivery risk incentivizes them to diversify their supplier base as a risk mitigation strategy. As a result, the number of sourcing relationships increases with aggregate risk, though it declines slightly when delays become almost certain and volatility is minimal. In contrast, risk-neutral firms maintain a relatively stable number of suppliers even as aggregate delivery risk increases. This divergence occurs because risk-neutral producers do not account for profit volatility in their optimization decisions, making the diversification benefits less attractive to them. Under our normalization where the sum of intermediate input weights in production equals one, adding additional suppliers does not expand the overall sourcing potential. Consequently, the cost reduction benefits from supplier diversification are limited. In this case, risk-neutral firms would optimally concentrate their sourcing on the single most cost-effective supplier, while risk-averse firms are willing to sacrifice some cost efficiency to reduce exposure to supply chain disruptions.

Figure 12 reveals the volatility patterns that highlight the effectiveness of risk-averse firms' diversification strategies. Under risk aversion (left panel), profit volatility maintain at relatively low level as delivery risk intensifies. This pattern reflects risk-averse firms' active portfolio management: as aggregate uncertainty increases, these firms respond by diversifying their supplier networks and shifting toward more reliable sources, effectively hedging against delivery shocks through strategic sourcing adjustments. In stark contrast, risk-neutral firms (right panel) show increasing volatility with aggregate risk, demonstrating their vulnerability to supply chain disruptions. These firms, focused solely on expected profits, maintain concentrated sourcing strategies that expose them to escalating uncertainty as delivery risks rise.

Figure 13,14,15 shows how the aggregate price level evolves as aggregate delivery risk rises. Across all levels of aggregate risk, risk-averse economies consistently exhibit higher expected aggregate prices than risk-neutral ones, demonstrating the price premium associated with risk mitigation strategies.

When firms are risk-averse, expected aggregate prices increase substantially with delivery risk, rising from approximately 0.18 to 0.23 as ρ_{agg} increases from 0 to 0.8. This price

[¶]the following simulation result run the version of code for 07132025

escalation reflects firms' willingness to pay premiums for supply chain stability. Crucially, this higher expected aggregate price stems from elevated prices in both good and bad states. Even when aggregate delays do not occur (AP Good), ex-post aggregate prices continue to rise with the aggregate delay probability ρ_{agg} and remain consistently higher than their risk-neutral counterparts, as producers pay premiums for maintaining diversified, reliable supply networks. This double price premium - higher costs in both good and bad states - demonstrates that uncertainty and the responsive risk mitigation is costly. Risk-averse firms essentially purchase "supply chain insurance" through their sourcing decisions, accepting permanently higher input costs in exchange for reduced profit volatility. The convergence of good-state and bad-state prices at $\rho_{agg} = 1$ reflects the elimination of uncertainty when disruptions become certain, reducing the value of diversification strategies.

Figure 16, 17, 18 illustrates the evolution of aggregate real income (E/P) across varying levels of delivery risk. While real income declines under both regimes as risk increases, expected aggregate real income remains consistently lower in the risk-averse case compared to the risk-neutral counterpart. This persistent gap reflects the economic costs of supply chain risk mitigation strategies. Importantly, the lower expected real income in risk-averse economies is primarily driven by reduced real income in the good state, when aggregate delays do not occur. Real income in good states falls substantially for risk-averse firms as delivery risk increases falling by approximately 17% from 1,450 at $\rho_{agg} = 0$ to around 1,200 at higher risk levels. In contrast, risk-neutral firms maintain stable real income in good states regardless of the underlying risk environment. This asymmetric pattern demonstrates that risk aversion imposes real economic costs even when disruptions do not materialize. Risk-averse firms sacrifice real income in favorable states to build resilience against potential supply chain disruptions, effectively purchasing "supply chain insurance" that reduces welfare even when it is not needed. The convergence of outcomes at $\rho_{agg} = 1$ reflects the elimination of uncertainty benefits when disruptions become certain.

Taken together, the simulations reveal a fundamental macroeconomic trade-off inherent in firms' precautionary sourcing strategies. While risk diversification behavior provides insurance against supply chain disruptions, these defensive strategies impose aggregate costs through higher equilibrium prices and reduced real income. Crucially, these welfare costs manifest even in the absence of actual disruptions, as risk-averse firms maintain costly diversified supplier networks based on the probability of future shocks. In economic environments characterized by high delivery risk, this analysis suggests that the equilibrium outcome features elevated prices and depressed output relative to low-risk environments, as firms collectively adopt precautionary measures that reduce efficiency while providing insurance value.

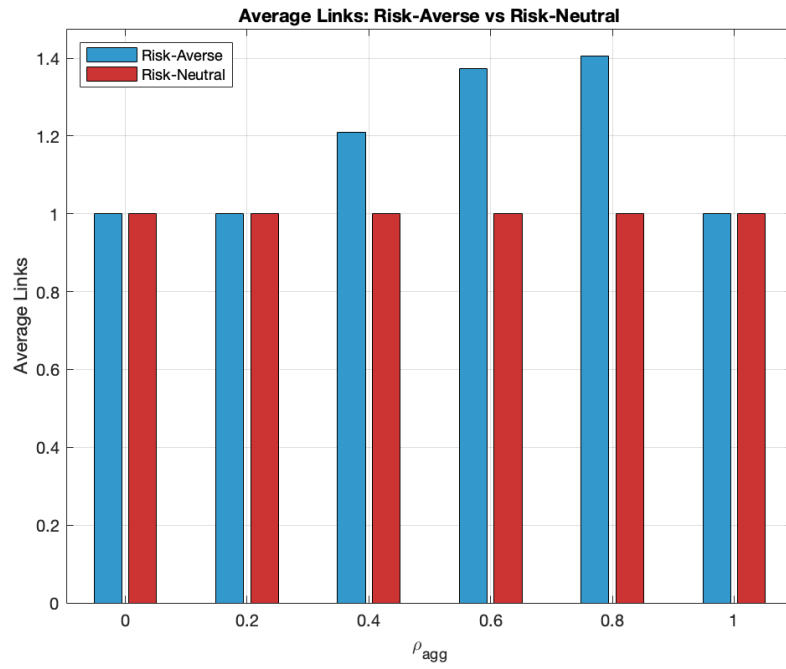


Figure 11: Average number of suppliers per producer

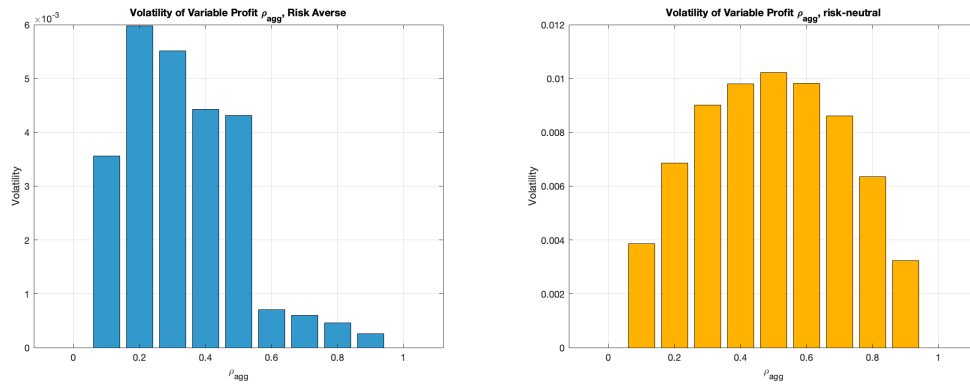


Figure 12: Volatility of variable profit

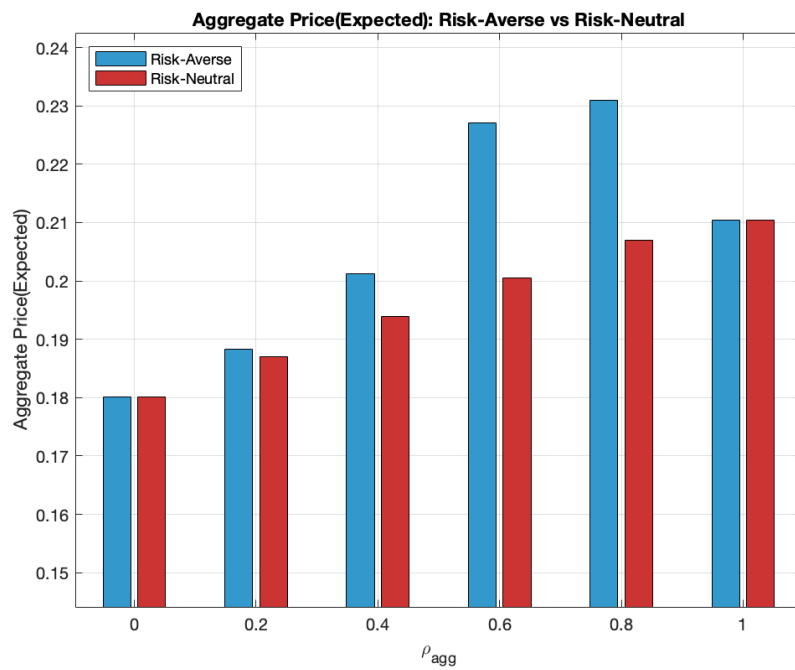


Figure 13: Aggregate Price, Risk Averse vs Risk Neutral

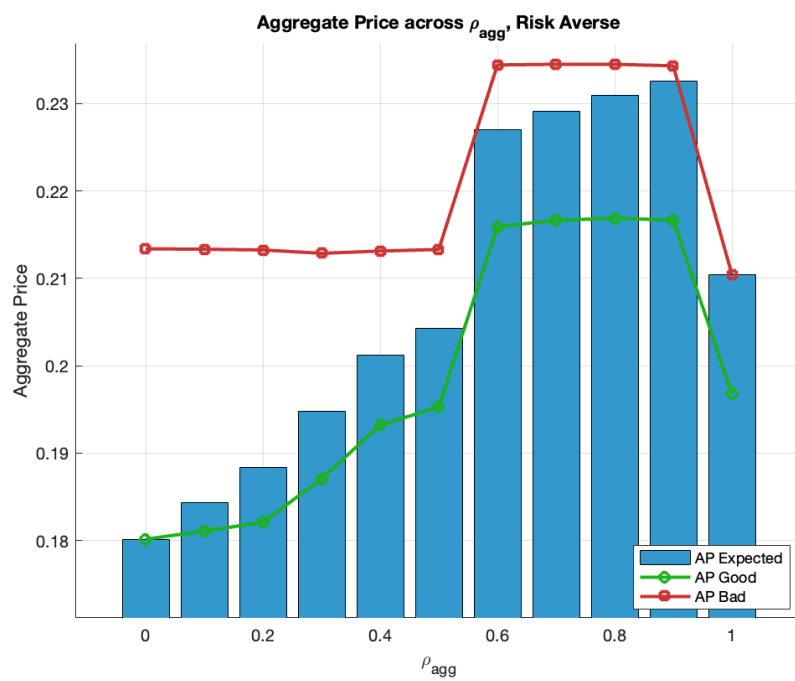


Figure 14: Aggregate Price, Risk Averse

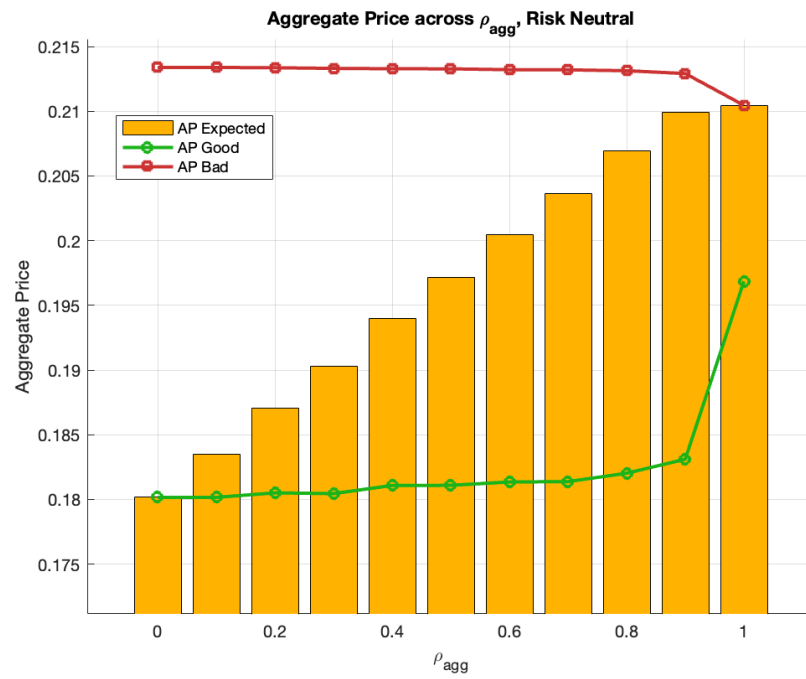


Figure 15: Aggregate Price, Risk Neutral

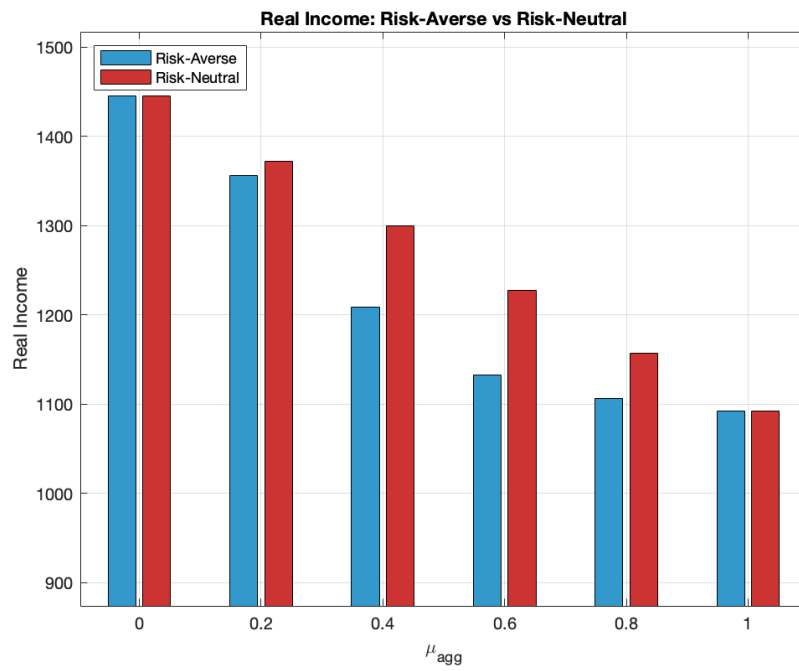


Figure 16: Real Income, Risk Averse vs Risk Neutral

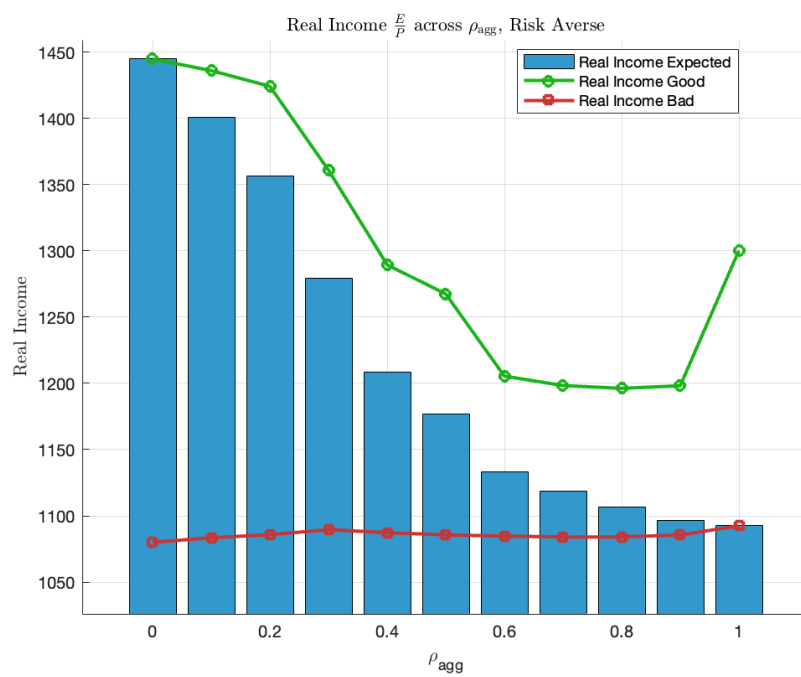


Figure 17: Real Income, Risk Averse

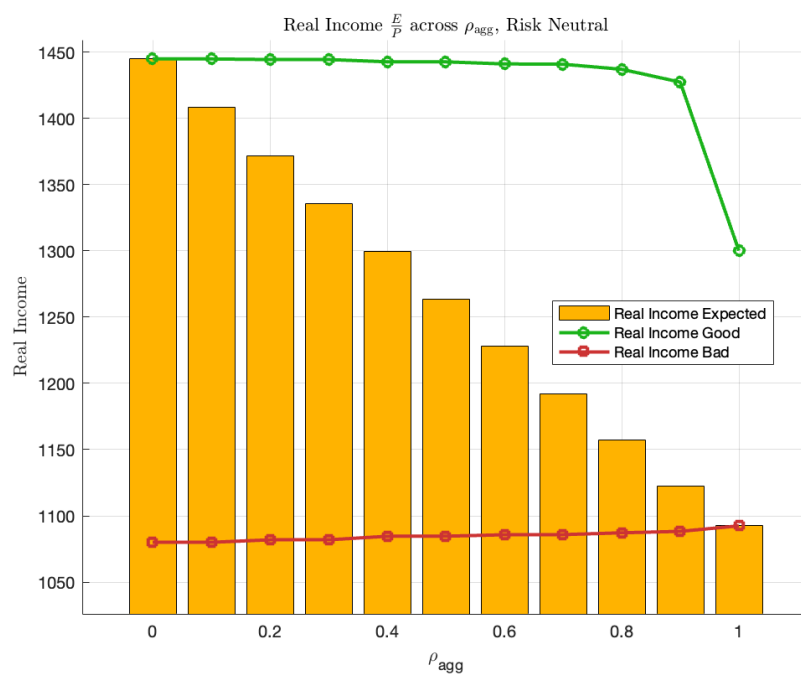


Figure 18: Real Income, Risk Neutral

6 Conclusion

This paper investigates how supply chain delivery risks reshape firm sourcing strategies and their macroeconomic consequences. Using granular supply chain relationship data, I document novel empirical evidence that firms systematically diversify their supplier networks in response to delivery uncertainty, shifting toward more reliable but less productive sources despite higher relationship-building costs. While this defensive sourcing strategy helps individual firms stabilize output and reduce profit volatility, it generates significant aggregate welfare costs through elevated equilibrium prices and reduced real income. To quantify these effects and understand their transmission mechanisms, I develop a general equilibrium model with heterogeneous firms facing endogenous supplier sourcing decisions under delivery risk. The simulation results show that firms display strong risk aversion in their sourcing choices, incorporating suppliers that are less productive but more reliable. Such precautionary strategies impose persistent macroeconomic costs, as economies that diversify to mitigate risk settle into equilibria with higher prices and lower income.

Crucially, these welfare losses manifest even in the absence of actual supply disruptions - simply the possibility of delivery delays drives costly precautionary behavior that reduces aggregate efficiency. The analysis reveals a fundamental trade-off between microeconomic stability and macroeconomic efficiency: while individual firms rationally insure against supply chain risks, their collective actions generate economy-wide costs that persist as long as underlying uncertainty remains elevated. These findings have important policy implications for understanding the economic costs of supply chain fragility. Policies aimed at reducing aggregate delivery uncertainty - through infrastructure investment, supply chain transparency initiatives, or coordination mechanisms - generate welfare benefits that extend far beyond crisis mitigation. By addressing the root causes of supply chain uncertainty, such interventions can eliminate the ongoing efficiency costs of precautionary behavior, supporting both price stability and long-term economic growth. In an era of increasing global supply chain complexity and geopolitical tensions, understanding and addressing these uncertainty driven inefficiencies becomes critical for maintaining economic resilience without sacrificing aggregate prosperity.

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A Alternative Model

A.1 Preferences and Demand

One representative consumer supplies given unit of labor inelastically to firms. Consumer has a CES preferences over differentiated consumption goods:

$$U = \left(\sum_{j \in \Omega} q_{jH}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (17)$$

where Ω denotes the set of final goods in the economy, k denotes a (differentiated or identical??) final goods producer, and H denotes the domestic final demand from households. The term β_{kH} captures how salient the good produced by firm k is in the final consumption bundle.

The budget constraint of the household is

$$E = \sum_{j \in \Omega} p_{jH} q_{jH} \quad (18)$$

where E denotes the aggregate expenditure.

The optimization problem of the household is

$$\max \left(\sum_{j \in \Omega} q_{jH}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \text{ st. } E = \sum_{j \in \Omega} p_{jH} q_{jH} \quad (19)$$

Standard algebra could derive following demand function that

$$q_{jH} = \frac{p_{jH}^{-\sigma}}{P^{1-\sigma}} E \quad (20)$$

where P denotes the domestic consumer price index:

$$P = \left(\sum_{j \in \Omega} p_{jH}^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \quad (21)$$

Detail algebra for demand function and price index see appendix B. We assume that final goods are substitutes, and therefore $\sigma > 1$.

A.2 Intermediate goods producer

Intermediate goods suppliers i only use labor as production input and sell it to final goods producer. The production function is $q_i = z_i \ell_i$, where z_i is their idiosyncratic productivity. The marginal cost of intermediates goods i is equal to $\frac{w}{z_i}$. p_{ij} denotes the price of intermediates good from intermediate supplier i to final goods producer j . We assume that intermediate firm i sets a price that is equal to an exogenous pair-specific markup times its marginal cost when selling to final good producer j , μ_{ij} . In the baseline case, we suppose that intermediate firms do not charge markups to final good producer (i.e. $\mu_{ij} = 1$). In this case, intermediate goods suppliers do not generate profits from their sales to final good producers.

Our model assume that intermediate suppliers have delivery uncertainty that there is probability that they suppliers could not deliver the input order placed by final good producers on time, considering the supply chain disruption.

A.3 Final good producers

Each final goods producer j produces a single differentiated product. All final goods producer use labor the intermediate goods source from intermediate goods suppliers to produce final goods and sell its own variety to final consumers.

Each final goods producer takes the marginal costs of its suppliers p_{ij} , the wage level w and aggregate expenditure E and consumer price index P as given. The final goods producer j follow the following production that

$$q_j = \phi_j \ell_j^\beta m_j^{1-\beta} \quad (22)$$

where ϕ_j denotes the exogenous TFP of final good producer j . β captures the importance of labor input for final goods producer's production. ℓ_j is final good producer's labor input and m_j is all intermediate input successfully delivered by all suppliers as perfect substitutes. We assume firm will use all intermediate input successfully delivered into production such that:

$$m_j(\mathbf{x}, \Delta) = \sum_{i \in Z_j} \Gamma_{ij} x_{ij}$$

Z_j denotes the set of intermediate good suppliers from which final good producer j is eligible to purchase inputs. x_{ij} is the order placed by final good producer to intermediate good supplier. $\Gamma_{ij} \in \{0, 1\}$ are random variable which indicate whether supplier i deliver to producer j successfully. $\Delta = (\Gamma_{1j}, \Gamma_{2j}, \dots, \Gamma_{Ij})^T \in \{0, 1\}^I$ is the vector of delivery in-

dicators for all suppliers. We assume a shock structure that the shocks between suppliers are correlated and the covariance matrix of the delivery indicators is denoted as $\Sigma \in \mathbb{R}^{I \times I}$. The off diagonal elements $\sigma_{jk} = \text{Cov}(\Gamma_j, \Gamma_k)$ captures the correlation due to shared delivery risks. For example, some suppliers may experience both geopolitical challenges and material shortage or natural disasters, while others may be affected by only one of these issues. As a result, delivery risks are correlated among suppliers.

Given a realization of Δ and the inverse demand function $p_j = \left(\frac{E}{P^{1-\sigma}}\right)^{\frac{1}{\sigma}} q_j^{-\frac{1}{\sigma}}$, the firm j 's profit is given by

$$\pi_j(\ell_j, \mathbf{x}, \Delta) = \left(\frac{E}{P^{1-\sigma}}\right)^{\frac{1}{\sigma}} q_j(\ell_j, \mathbf{x}, \Delta)^{1-\frac{1}{\sigma}} - \sum_{i \in Z_j} p_{ij} x_{ij} - w \ell_j \quad (23)$$

The expected utility function of the firm is

$$U(\mathbf{q}) = \mathbb{E}_\Delta[\pi_j(\ell_j, \mathbf{x}, \Delta)] - \lambda \cdot \text{Var}_\Delta(\pi_j(\ell_j, \mathbf{x}, \Delta)) \quad (24)$$

Currently, we assume firm is risk averse, governed by λ , so that volatility of the profit will affect firm's utility. The optimization problem of the final good producer is that, prior to the realization of the delivery shock, takes the marginal cost of its suppliers p_{ij} , the wage level w , the aggregate E and consumer price index P as given, final good producers choose their orders x_{ij} and labor input l_j to maximize the expected utility. Firms have rational expectations and make their input sourcing based on the true joint distribution of the delivery risk. Once the sourcing decision is made ex-ante, production outcomes and profits are automatically determined as shocks materialize under each state. The optimization aims to find the sourcing strategy that maximizes expected utility, taking delivery risk into account.

The assumption of firm risk aversion contradicts the common assumption that firms are risk-neutral. Here, however, we haven't explicitly modeled demand; we simply assume that households will purchase everything produced. In future research, if we assume households are risk-averse and that firms are owned by households, it may be possible to achieve diversification under the assumption of risk-neutral firms.

Additionally, in paper Castro-Vincenzi et al. 2024, climate risk is modeled similarly, with a probability that initial orders may not be fulfilled in a spatial model. However, a key difference is that they choose labor and production quantity after the shock is realized, whereas here, labor is chosen ex-ante. In their two-stage model, the expected profit function is concave, making the firm effectively risk-averse without an explicit risk-aversion assumption. Exploring this approach further is a next step I am interested in pursuing.

A.4 A Three Suppliers Example

To build intuition, consider a scenario in the electric vehicle industry with three suppliers, each varying in productivity and delivery risk. Assume the first supplier is a Chinese lithium battery company facing delivery risks due to political issues and lithium shortages. The second supplier, a Chinese lead provider, also faces political risks but has no material shortage concerns. The third supplier is a South Korean lithium supplier with delivery risks stemming from material shortages. Suppose the productivity levels of these suppliers are such that $z_1 > z_2 > z_3$. In this setup, delivery risks are correlated between Supplier 1 and Supplier 2, as well as between Supplier 1 and Supplier 3, but are independent between Supplier 2 and Supplier 3. This creates a trade-off between productivity and delivery risk, with the most productive supplier being the most vulnerable to delivery risks.

I haven't fully solved the model yet. Below is only the firm-side optimization, where price is represented as a reduced-form equation based on realized production, ie $p_j = (\frac{E}{p^{1-\sigma}})^{\frac{1}{\sigma}} q_j^{-1/\sigma}$ and $(\frac{E}{p^{1-\sigma}})^{\frac{1}{\sigma}}$ is exogenous here. In order to find the equilibrium, I need to simulate the model and finding the fixed point of demand shifter $\frac{E}{p^{1-\sigma}}$ state by state, which I will do in the future.

Figure 19 The results show the sourcing strategy from three suppliers and the unit cost of supply, defined as the total cost the producer pays for input divided by the sum of input quantities. These metrics illustrate how sourcing patterns and costs shift with changes in delivery risk. When risk is low, the producer sources from two Chinese suppliers who are productive but carry higher risk. As the risk increases, the producer begins to diversify by sourcing from the third supplier, who is less productive but safer. Our risk diversification strategy demonstrates that producers diversify not only among suppliers within the same industry but also across different industries. If the risk continues to rise, the producer abandons the most risky supplier and relies on the two safer options. During the diversification phase, the input unit cost (total order cost divided by order quantity) rises sharply as the producer includes the most expensive supplier. Additionally, the total input ordered declines, signaling a reduction in total production, even as the supply chain becomes more secure.

Figure 20 show that the expected production follows a convex curve, decreasing at a slower rate as risk increases after diversification. Figure 21 illustrates the trade-off between expected profit and input delivery volatility. While the diversification strategy effectively reduces input volatility, it also results in lower expected profit during this phase.

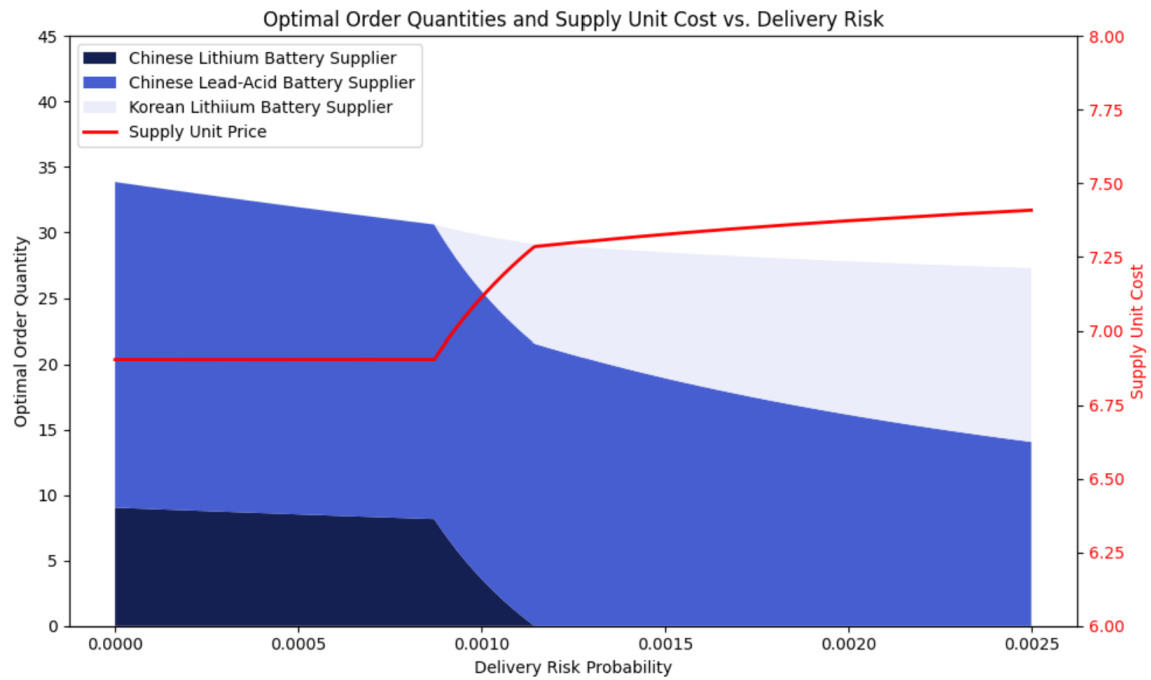


Figure 19: Supplier Sourcing

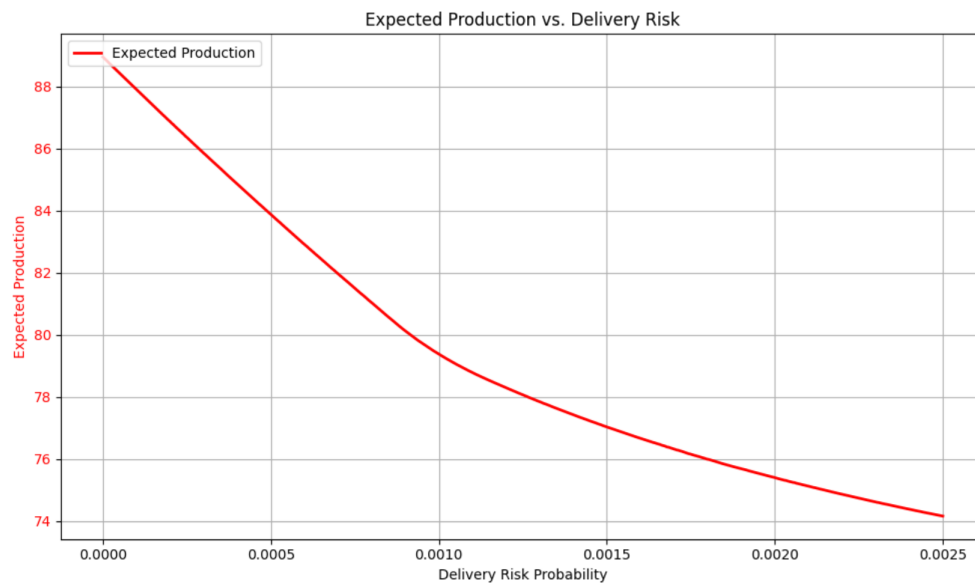


Figure 20: Expected Cost and Product



Figure 21: Expected Profit and Volatility

B Appendix: Demand Function and Price Index

$$\max \left(\sum_{j \in \Omega} (\beta_{jH} q_{jH})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \text{st. } E = \sum_{j \in \Omega} p_{jH} q_{jH}$$

Lagrangian

$$\mathcal{L} = \left(\sum_{j \in \Omega} (\beta_{jH} q_{jH})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} + \lambda \left(E - \sum_{j \in \Omega} p_{jH} q_{jH} \right)$$

$$\frac{\partial \mathcal{L}}{\partial q_{iH}} = \frac{\sigma}{\sigma-1} \left(\sum_{j \in \Omega} (\beta_{jH} q_{jH})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} \frac{\sigma-1}{\sigma} (\beta_{iH} q_{iH})^{-\frac{1}{\sigma}} \beta_{iH} - \lambda p_{iH} = 0 \quad (25)$$

$$\frac{\partial \mathcal{L}}{\partial q_{sH}} = \frac{\sigma}{\sigma-1} \left(\sum_{j \in \Omega} (\beta_{jH} q_{jH})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} \frac{\sigma-1}{\sigma} (\beta_{sH} q_{sH})^{-\frac{1}{\sigma}} \beta_{sH} - \lambda p_{sH} = 0 \quad (26)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = E - \sum_{j \in \Omega} p_{jH} q_{jH} = 0 \quad (27)$$

Divide (1) to (2)

$$\frac{p_{iH}}{p_{sH}} = \frac{(\beta_{iH})^{\frac{\sigma-1}{\sigma}} q_{iH}^{-\frac{1}{\sigma}}}{(\beta_{sH})^{\frac{\sigma-1}{\sigma}} q_{sH}^{-\frac{1}{\sigma}}}$$

Solve for q_{iH}

$$\frac{q_{sH}}{q_{iH}} = \left(\frac{p_{sH}}{p_{iH}} \right)^{\sigma} \left(\frac{\beta_{sH}}{\beta_{iH}} \right)^{1-\sigma}$$

$$q_{sH} = \left(\frac{p_{iH}}{p_{sH}} \right)^{\sigma} \left(\frac{\beta_{iH}}{\beta_{sH}} \right)^{1-\sigma} q_{iH} \quad (28)$$

Replace all q_{jH} variables in $E = \sum_{j \in \Omega} p_{jH} q_{jH}$ as a function of q_{iH}

Substitute q_{sH} into $E = \sum_{j \in \Omega} p_{jH} q_{jH}$

$$\begin{aligned} E &= p_{iH} q_{iH} + \sum_{s \neq i} p_{sH} q_{sH} \\ &= p_{iH} q_{iH} + \sum_{s \neq i} p_{sH} \left(\frac{p_{iH}}{p_{sH}} \right)^{\sigma} \left(\frac{\beta_{iH}}{\beta_{sH}} \right)^{1-\sigma} q_{iH} \\ &= q_{iH} \left(p_{iH} + \sum_{s \neq i} p_{sH}^{1-\sigma} p_{iH}^{\sigma} \left(\frac{\beta_{iH}}{\beta_{sH}} \right)^{1-\sigma} \right) \\ &= q_{iH} \left(p_{iH}^{1-\sigma} p_{iH}^{\sigma} \left(\frac{\beta_{iH}}{\beta_{iH}} \right)^{1-\sigma} + \sum_{s \neq i} p_{sH}^{1-\sigma} p_{iH}^{\sigma} \left(\frac{\beta_{iH}}{\beta_{sH}} \right)^{1-\sigma} \right) \\ &= q_{iH} \left(\sum_{j \in \Omega} p_{jH}^{1-\sigma} p_{iH}^{\sigma} \left(\frac{\beta_{jH}}{\beta_{iH}} \right)^{\sigma-1} \right) \end{aligned}$$

Now we can solve q_{iH} as a function of $\beta_{kH}, \beta_{iH}, p_{kH}, p_{iH}$ and E

$$q_{iH} = \frac{E}{\sum_{k \in \Omega} \left(\frac{\beta_{jH}}{\beta_{iH}} \right)^{\sigma-1} p_{jH}^{1-\sigma} p_{iH}^{\sigma}}$$

Derive the aggregate price P

Substitute $q_{iH} = \frac{E}{\sum_{j \in \Omega} \left(\frac{\beta_{jH}}{\beta_{iH}} \right)^{\sigma-1} p_{jH}^{1-\sigma} p_{iH}^{\sigma}}$ into $Q = \left(\sum_{j \in \Omega} (\beta_{jH} q_{jH})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$

$$\begin{aligned}
Q &= \left(\sum_{i \in \Omega} \left(\beta_{iH} \frac{E}{\sum_{j \in \Omega} \left(\frac{\beta_{jH}}{\beta_{iH}} \right)^{\sigma-1} p_{jH}^{1-\sigma} p_{iH}^{\sigma}} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \\
&= \left(\sum_{i \in \Omega} \left(\frac{E (\beta_{iH})^{\sigma} p_{iH}^{-\sigma}}{\sum_{j \in \Omega} (\beta_{jH})^{\sigma-1} p_{jH}^{1-\sigma}} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \\
&= E \left(\sum_{j \in \Omega} (\beta_{jH})^{\sigma-1} p_{jH}^{1-\sigma} \right)^{-1} \left(\sum_{i \in \Omega} (\beta_{iH})^{\sigma-1} p_{iH}^{1-\sigma} \right)^{\frac{\sigma}{\sigma-1}} \\
&= E \left(\sum_{j \in \Omega} (\beta_{jH})^{\sigma-1} p_{jH}^{1-\sigma} \right)^{\frac{1}{\sigma-1}}
\end{aligned}$$

Since $E = P * Q$, we have

$$P = \left(\sum_{j \in \Omega} (\beta_{jH})^{\sigma-1} p_{jH}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$$

Substitute P into q_{iH}

$$q_{iH} = \frac{E}{\sum_{j \in \Omega} \left(\frac{\beta_{jH}}{\beta_{iH}} \right)^{\sigma-1} p_{jH}^{1-\sigma} p_{iH}^{\sigma}} = \frac{E p_{iH}^{-\sigma} (\beta_{iH})^{\sigma-1}}{\sum_{j \in \Omega} (\beta_{jH})^{\sigma-1} p_{jH}^{1-\sigma}} = (\beta_{iH})^{\sigma-1} \frac{p_{iH}^{-\sigma}}{p^{1-\sigma}} E$$

C Appendix: Cost function

$$\min \sum_{i \in Z_j} p_{ij} q_{ij} + w l_j$$

$$\text{st. } q_j = \phi_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij} q_{ij})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj} \ell_j)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

Lagrangian

$$\mathcal{L} = \sum_{i \in Z_j} p_{ij} q_{ij} + w l_j + \lambda \left[\phi_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij} q_{ij})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj} w)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} - q_j \right]$$

$$\frac{\partial L}{\partial q_{ij}} = p_{ij} + \lambda \frac{\sigma}{\sigma-1} \phi_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij} q_{ij})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj} w)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} \frac{\sigma-1}{\sigma} (\alpha_{ij} \Gamma_{ij})^{\frac{\sigma-1}{\sigma}} q_{ij}^{-\frac{1}{\sigma}} = 0 \quad (29)$$

$$\frac{\partial L}{\partial q_{kj}} = p_{kj} + \lambda \frac{\sigma}{\sigma-1} \phi_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij} q_{ij})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj} w)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} \frac{\sigma-1}{\sigma} (\alpha_{kj} \Gamma_{kj})^{\frac{\sigma-1}{\sigma}} q_{kj}^{-\frac{1}{\sigma}} = 0 \quad (30)$$

$$\frac{\partial L}{\partial l_j} = w + \lambda \frac{\sigma}{\sigma-1} \phi_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij} q_{ij})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj} w)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} \frac{\sigma-1}{\sigma} \alpha_{Lj}^{\frac{\sigma-1}{\sigma}} l_j^{-\frac{1}{\sigma}} = 0 \quad (31)$$

Divide xx by xx, we have

$$\frac{p_{kj}}{p_{ij}} = \left(\frac{\alpha_{kj} \Gamma_{kj}}{\alpha_{ij} \Gamma_{ij}} \right)^{\frac{\sigma-1}{\sigma}} \left(\frac{q_{kj}}{q_{ij}} \right)^{-\frac{1}{\sigma}}$$

$$\frac{w}{p_{ij}} = \left(\frac{\alpha_{Lj}}{\alpha_{ij} \Gamma_{ij}} \right)^{\frac{\sigma-1}{\sigma}} \left(\frac{l_j}{q_{ij}} \right)^{-\frac{1}{\sigma}}$$

Thus

$$q_{kj} = \left(\frac{p_{kj}}{p_{ij}} \right)^{-\sigma} \left(\frac{\alpha_{kj} \Gamma_{kj}}{\alpha_{ij} \Gamma_{ij}} \right)^{\sigma-1} q_{ij}$$

$$l_j = \left(\frac{w}{p_{ij}} \right)^{-\sigma} \left(\frac{\alpha_{Lj}}{\alpha_{ij} \Gamma_{ij}} \right)^{\sigma-1} q_{ij}$$

Substitute derived q_{kj} and l_j into $C_j = \sum_{k \in Z_j} p_{kj} q_{kj} + w l_j$

we have

$$C_j = \sum_{k \in Z_j} p_{kj} \left(\frac{p_{kj}}{p_{ij}} \right)^{-\sigma} \left(\frac{\alpha_{kj} \Gamma_{kj}}{\alpha_{ij} \Gamma_{ij}} \right)^{\sigma-1} q_{ij} + w \left(\frac{w}{p_{ij}} \right)^{-\sigma} \left(\frac{\alpha_{Lj}}{\alpha_{ij} \Gamma_{ij}} \right)^{\sigma-1} q_{ij} \quad (32)$$

$$= q_{ij} \frac{p_{ij}^{\sigma}}{(\alpha_{ij} \Gamma_{ij})^{\sigma-1}} \left(\sum_{k \in Z_j} p_{kj}^{-\sigma+1} (\alpha_{kj} \Gamma_{kj})^{\sigma-1} + w^{-\sigma+1} \alpha_{Lj}^{\sigma-1} \right) \quad (33)$$

Rearrange, we have

$$q_{ij} = \frac{C_j}{\left(\sum_{k \in Z_j} p_{kj}^{-\sigma+1} (\alpha_{kj} \Gamma_{kj})^{\sigma-1} + w^{-\sigma+1} \alpha_{Lj}^{\sigma-1} \right)} \frac{(\alpha_{ij} \Gamma_{ij})^{\sigma-1}}{p_{ij}^{\sigma}}$$

Similarly, we have

$$\ell_j = \frac{C_j}{\left(\sum_{k \in Z_j} p_{kj}^{-\sigma+1} (\alpha_{kj} \Gamma_{kj})^{\sigma-1} + w^{-\sigma+1} \alpha_{Lj}^{\sigma-1} \right)} \frac{\alpha_{Lj}^{\sigma-1}}{w^{\sigma}}$$

Denote $\Theta_j = \sum_{k \in Z_j} p_{kj}^{-\sigma+1} (\alpha_{kj} \Gamma_{kj})^{\sigma-1} + w^{-\sigma+1} \alpha_{Lj}^{\sigma-1}$ and substitute $q_{ij} = \frac{C_j}{\Theta_j} \frac{(\alpha_{ij} \Gamma_{ij})^{\sigma-1}}{p_{ij}^{\sigma}}$, $\ell_j = \frac{C_j}{\Theta_j} \frac{\alpha_{Lj}^{\sigma-1}}{w^{\sigma}}$ into $q_j = \phi_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij} q_{ij})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj} \ell_j)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$

We have

$$q_j = \phi_j \left(\sum_{i \in Z_j} \left(\alpha_{ij} \Gamma_{ij} \frac{C_j}{\Theta_j} \frac{(\alpha_{ij} \Gamma_{ij})^{\sigma-1}}{p_{ij}^{\sigma}} \right)^{\frac{\sigma-1}{\sigma}} + \left(\alpha_{Lj} \frac{C_j}{\Theta_j} \frac{\alpha_{Lj}^{\sigma-1}}{w^{\sigma}} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (34)$$

$$= \phi_j \frac{C_j}{\Theta_j} \left(\sum_{i \in Z_j} ((\alpha_{ij} \Gamma_{ij})^{\sigma} p_{ij}^{-\sigma})^{\frac{\sigma-1}{\sigma}} + (\alpha_{Lj}^{\sigma} w^{-\sigma})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (35)$$

$$= \phi_j \frac{C_j}{\Theta_j} \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma} \right)^{\frac{\sigma}{\sigma-1}} \quad (36)$$

$$= \phi_j C_j \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma} \right)^{\frac{1}{\sigma-1}} \quad (37)$$

$$= \phi_j C_j \Theta_j^{\frac{1}{\sigma-1}} \quad (38)$$

where $\Theta_j = \sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma}$

Finally, we get

$$c_j = \frac{C_j}{q_j} = \frac{1}{\phi_j} \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma} \right)^{\frac{1}{1-\sigma}} = \frac{1}{\phi_j} \Theta_j^{\frac{1}{1-\sigma}}$$

D Appendix Profit function

Write profit function as a function of unit variable cost c_j

$$\pi_{jH} = x_{jH} - c_j q_j$$

Given the preference, we can derive the demand function that

$$q_j = (\beta_{jH})^{\sigma-1} p_{jH}^{-\sigma} \frac{E}{P^{1-\sigma}}$$

Note that

$$x_{jH} = p_{jH} q_j = p_{jH}^{1-\sigma} (\beta_{jH})^{\sigma-1} \frac{E}{P^{1-\sigma}} = (\mu c_{jH})^{1-\sigma} (\beta_{jH})^{\sigma-1} \frac{E}{P^{1-\sigma}}$$

Thus, express profit as a function of c_j

$$\begin{aligned} \pi_{jH} &= x_{jH} - c_j q_j = (p_{jH} - c_j) q_j \\ &= (\mu - 1) c_j (\beta_{jH})^{\sigma-1} p_{jH}^{-\sigma} \frac{E}{P^{1-\sigma}} \\ &= (\mu - 1) c_j (\beta_{jH})^{\sigma-1} (\mu c_j)^{-\sigma} \frac{E}{P^{1-\sigma}} \\ &= c_j^{1-\sigma} \left(1 - \frac{1}{\mu}\right) \mu^{1-\sigma} (\beta_{jH})^{\sigma-1} \frac{E}{P^{1-\sigma}} \end{aligned}$$

Express profit as a function of p_{jH}

$$\begin{aligned} \pi_{jH} &= x_{jH} - c_j q_j = (p_{jH} - c_j) q_j \\ &= (\mu - 1) c_j \beta_{jH}^{\sigma-1} p_{jH}^{-\sigma} \frac{E}{P^{1-\sigma}} \\ &= (\mu - 1) \frac{p_{jH}}{\mu} \beta_{jH}^{\sigma-1} (p_{jH})^{-\sigma} \frac{E}{P^{1-\sigma}} \\ &= p_{jH}^{1-\sigma} \left(1 - \frac{1}{\mu}\right) \beta_{jH}^{\sigma-1} \frac{E}{P^{1-\sigma}} \end{aligned}$$

Derive the implied expected unit variable cost and implied price when maximize expected profit

$$c_j = \frac{C_j}{q_j} = \frac{1}{\phi_j} \left(\sum_{i \in Z_j} (\alpha_{ij} \Gamma_{ij})^{\sigma-1} p_{ij}^{1-\sigma} + \alpha_{Lj}^{\sigma-1} w^{1-\sigma} \right)^{\frac{1}{1-\sigma}} = \frac{1}{\phi_j} \Theta_j(\Gamma)_{ij}^{\frac{1}{1-\sigma}}$$

$$\begin{aligned} E(\pi_{jH}) &= E(c_j)^{1-\sigma} \left(1 - \frac{1}{\mu}\right) \mu^{1-\sigma} \beta_{jH}^{\sigma-1} \frac{E}{P^{1-\sigma}} \\ &= E(p_{jH})^{1-\sigma} \left(1 - \frac{1}{\mu}\right) \beta_{jH}^{\sigma-1} \frac{E}{P^{1-\sigma}} \end{aligned}$$

where $E(p_{jH}) = \mu E(c_j)$

$$E(\theta_j) = E(\phi_j c_j)^{1-\sigma} = \phi_j^{1-\sigma} E(c_j)^{1-\sigma}$$